



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

BE – Makeup Exam

I to II Semester

Jun/Jul-2024

(2022 Scheme)

VVIET

Bannur road, Alanahalli, Mysore-570028

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CBCS SCHEME

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BBEE103/203

**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Basic Electronics for EEE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the forward and reverse characteristics of a silicon diode.	08	L2	CO1
	b.	Describe the working of a two diode full wave rectifier with neat circuit diagram and waveforms.	08	L2	CO1
	c.	Explain the working of a capacitor filter for a half wave rectifier with neat circuit diagram and waveforms.	04	L2	CO1
OR					
Q.2	a.	Explain the working of a full wave bridge rectifier with neat circuit diagram and waveforms.	08	L2	CO1
	b.	Illustrate how Zener diode can be used as voltage regulator in load and no load conditions.	08	L2	CO1
	c.	Write a note on ideal, practical and piecewise linear characteristics of a pn junction diode.	04	L2	CO1
Module – 2					
Q.3	a.	Explain the input and output characteristics of a transistor in common Emitter configuration.	08	L2	CO2
	b.	Indicate the various voltages and currents in a transistor circuit and derive β_{dc} in terms of α_{dc} .	06	L2	CO2
	c.	Explain the operating principle of an n channel JFET.	06	L2	CO2
OR					
Q.4	a.	Explain the working of an n-channel enhancement type MOSFET.	08	L2	CO2
	b.	Describe the input and output characteristics of a transistor connected in common base configuration.	06	L2	CO1
	c.	Explain how the DC load line can be constructed with respect to a BJT.	06	L2	CO1
Module – 3					
Q.5	a.	Define the following with respect to an opamp: (i) Slew rate (ii) CMRR (iii) Input Offset Voltage	06	L2	CO1
	b.	Explain how an opamp can be used as an integrator and differentiator.	08	L2	CO2
	c.	An opamp inverting amplifier has a feedback resistor of 15 K Ω and a input resistor of 2K Ω . Calculate the gain of the opamp and the output voltage if it is supplied with an input of 0.75V.	06	L3	CO2
OR					
Q.6	a.	List the characteristics of an ideal opamp.	06	L2	CO1
	b.	Describe how an opamp can be used as a difference amplifier/subtracting circuit.	08	L2	CO2
	c.	Show how an opamp summing circuit can be used to produce an output voltage: $V_o = -(3V_1 + 3V_2 + 3V_3)$	06	L3	CO2

Module – 4

Q.7	a.	Perform binary subtraction using 2's complement method on the following: (i) $(1101)_2 - (1010)_2$ (ii) $(10101.101)_2 - (01110.111)_2$	04	L3	CO3
	b.	Perform the following conversions: (i) $(82.25)_{10} = (?)_2$ (ii) $(46)_{10} = (?)_8$ (iii) $(126)_{10} = (?)_{16}$ (iv) $(111011.1001)_2 = (?)_{10}$	08	L3	CO3
	c.	Simplify the following Boolean functions: i) $(A + \overline{B}C)(\overline{A} + B + \overline{C})(A + \overline{B})$ ii) $A + A\overline{B} + \overline{A}B$	08	L3	CO3

OR

Q.8	a.	With neat diagram, explain Half adder.	02	L2	CO3
	b.	Explain how a full adder can be realized using two half adders.	08	L2	CO3
	c.	List the Boolean laws. Also state and prove Demorgan's theorem.	10	L2	CO3

Module – 5

Q.9	a.	Describe the working of a LVDT with neat diagram.	08	L2	CO5
	b.	Explain the general block diagram of communication system.	06	L2	CO4
	c.	Define modulation. What are the needs of modulation.	06	L2	CO4

OR

Q.10	a.	Explain the working of potentiometer-type resistive transducer.	08	L2	CO5
	b.	Describe piezo electric transducer.	06	L2	CO5
	c.	Write a short note on photodiodes.	06	L2	CO5

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**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Applied Chemistry for Civil Engineering Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	What is glass? Describe the preparation of soda lime glass.	07	L2	CO1
	b.	Explain the testing of element by EDTA method.	07	L2	CO1
	c.	Write the properties and applications of stainless steel and duralumin.	06	L2	CO1
OR					
Q.2	a.	Discuss the production of cement by wet process.	07	L2	CO1
	b.	Define Refractories. Mention the properties and applications of refractory materials.	07	L2	CO1
	c.	Explain the properties and applications of Aluminium and its alloys.	06	L2	CO1
Module – 2					
Q.3	a.	Explain the electrochemical corrosion of steel in concrete.	07	L2	CO2
	b.	Illustrate the construction and working of photovoltaic cell.	07	L2	CO2
	c.	Discuss the following type of corrosion: (i) Differential metal corrosion (ii) Differential aeration corrosion	06	L2	CO2
OR					
Q.4	a.	Define secondary batteries. Explain construction and working of Li-ion battery.	07	L2	CO2
	b.	What is anodizing? Explain the anodizing of aluminium.	07	L2	CO2
	c.	Explain the construction and working of Methanol-Oxygen fuel cell.	06	L2	CO2
Module – 3					
Q.5	a.	100 ml of sample water required 20 ml of 0.01 M EDTA for titration using EBT indicator. In another experiment 100 ml of the sample of water was boiled and precipitate was removed by filtration required 8.0 ml of 0.01 M EDTA using EBT indicator. Calculate: (i) Total Hardness (ii) Permanent Hardness (iii) Temporary Hardness	07	L2	CO3
	b.	What is desalination? Describe the desalination of water by electrodialysis method with a neat labelled diagram.	07	L2	CO3
	c.	Explain the following size dependent properties of nanomaterials: (i) Surface area (ii) Catalytic property	06	L2	CO3
OR					
Q.6	a.	What are hard waters? Discuss the estimation of total hardness of using EDTA solution.	07	L2	CO3
	b.	Define COD. In a COD test, 28 cm ³ and 16 cm ³ of 0.05 N FAS solutions are required for a blank and sample titration respectively. The volume of sample used was 25 cm ³ . Find the COD of a sample solution.	07	L2	CO3
	c.	Define nanomaterial. Explain the synthesis of nanomaterials by sol-gel method.	06	L2	CO3

Module – 4

Q.7	a.	Describe the synthesis, properties and applications of polyethylene.	07	L2	CO4
	b.	Define biodegradable polymer. Explain the synthesis and applications of polylactic acid.	07	L2	CO4
	c.	What are polymer composites? Mention the properties and application of FRP (Fiber Reinforced Polymer).	06	L2	CO4

OR

Q.8	a.	What are adhesives? Give the synthesis, properties and applications of epoxy resin.	07	L2	CO4
	b.	A polymer sample contains 50 molecules of molecular mass 1000 g/mols, 100 molecules of molecular mass 2000 g/mols and 150 molecules of molecular mass 3000 g/mols. Calculate the number average and weight average molecular masses of the polymer.	07	L3	CO4
	c.	Mention the properties and applications of geopolymer concrete.	06	L2	CO4

Module – 5

Q.9	a.	Describe the Lead-Silver system with the help of neat phase diagram.	07	L2	CO5
	b.	Describe the instrumentation and applications of conductometric sensors in the estimation of acid mixture.	07	L2	CO5
	c.	Describe the estimation of FAS using potentiometric sensors.	06	L2	CO5

OR

Q.10	a.	Illustrate the principle, instrumentation and working of potentiometric sensors.	07	L2	CO5
	b.	Define the following terms with examples: (i) Phase (ii) Component (iii) Degree of freedom	07	L2	CO5
	c.	Explain the determination of p^H of soil sample using p^H sensors.	06	L2	CO5

CBCS SCHEME

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BCHEE102/202

**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Chemistry for EEE Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Explain purification of electronic grade si using quartz by float zone method.	07	L2	CO1
	b.	Describe electroless plating of cu in the manufacture of PCB.	07	L2	CO1
	c.	Define : (i) Conducting polymers (ii) Number average molecular weight (iii) Weight average molecular weight	06	L2	CO1
OR					
Q.2	a.	Explain classification of materials as conductors, semiconductors and insulators using band theory.	07	L2	CO1
	b.	Explain synthesis, properties and applications of graphene oxide.	06	L2	CO1
	c.	In a polymer sample, 20% of molecules have molecular mass 15000 g/mol, 45% molecules have molecular mass 25000 g/mol, remaining molecules have molecular mass 27000 g/mol. Calculate number average mol.wt (M_n) and weight average molecular weight (M_w).	07	L3	CO1
Module – 2					
Q.3	a.	Explain construction and working of Na-ion battery and mention its application.	07	L2	CO2
	b.	What are fuel cells? Explain construction and working of Methanol – O ₂ fuel cell.	07	L2	CO2
	c.	Explain construction and working of PV cells.	06	L2	CO2
OR					
Q.4	a.	What are batteries? Explain classification of batteries with examples.	07	L2	CO2
	b.	Explain the construction and working of Lithium-polymer battery. Mention its applications.	07	L2	CO2
	c.	Explain construction and working of polymer electrolyte membrane fuel cell.	06	L2	CO2
Module – 3					
Q.5	a.	Define corrosion. Explain electrochemical theory of corrosion taking Fe as example.	07	L2	CO3
	b.	Explain the process of galvanization with applications.	07	L2	CO3
	c.	Explain sacrificial anodic method of prevention of corrosion.	06	L2	CO3
OR					
Q.6	a.	What is Anodising? Explain Anodising of Al.	07	L2	CO3
	b.	What is e-waste? Explain the methods of e-waste disposal.	06	L2	CO3
	c.	A sheet of carbon steel meter wide by three meters long has lost 40g to corrosion over the past six months. Calculate corrosion penetration rate (cpr) in mpy and mmpy. (carbon steel density = 7.8 g/cc, K = 87.6 mmpy, K = 534 mpy)	07	L3	CO3

Module – 4

Q.7	a.	What are Nano materials? Explain any two size dependent properties of nano materials.	07	L2	CO4
	b.	Explain synthesis of nano materials by sol gel method.	06	L2	CO4
	c.	What are liquid crystals? Explain the properties and applications of OLED and QLED.	07	L2	CO4

OR

Q.8	a.	What are pervoskite materials? Give the properties and applications of pervoskite materials.	07	L1	CO4
	b.	Write a note on nanofibres and nano sensors.	06	L1	CO4
	c.	Explain the classification of liquid crystals. Mention their application.	07	L2	CO4

Module – 5

Q.9	a.	What are reference electrodes? Explain construction and working of calomel electrode.	07	L2	CO5
	b.	Explain the process of determination of p^H of Vinegar using glass electrode.	06	L2	CO5
	c.	Represent a cell formed by immersing two silver electrodes in $AgNO_3$ solution of concentration 0.01 and 0.1 M. Write the reactions and find the emf of the cell.	07	L3	CO5

OR

Q.10	a.	Explain principle, instrumentation and application of potentiometric sensors in estimation of iron.	07	L3	CO5
	b.	Explain how the strength of a weak acid is determined using a conductometric sensor.	07	L2	CO5
	c.	Explain how cu is estimated using colorimeter.	06	L3	CO5

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BCHEM102/202

First/Second Semester B.E./B.Tech. Degree Supplementary Examination, June/July 2024

Applied Chemistry for ME Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define Chemical fuel. Explain the classification of fuel with examples.	7	L1,2	CO1
	b.	Define GCV and NCV. Explain the determination of calorific values of a solid fuel using bomb calorimeter.	7	L3	CO1
	c.	The 0.85g of coal sample (carbon = 90% H ₂ = 5% and ash = 5%) was subjected to combustion in a bomb calorimeter mass of water taken in the calorimeter was 2000g and the water equivalent of the calorimeter was 600g. The rise in temperature was found to be 3.5°C. Calculate the gross and net calorific values of the sample. (Latent heat of steam = 2.454kJ/g ; specific heat of water = 4.2kJ/kg °C)	6	L4	CO1
OR					
Q.2	a.	What is Fuel cell? Explain the construction and working of Methanol – Oxygen fuel cell.	7	L3	CO1
	b.	Explain the construction, working and application of PV cell.	6	L3	CO1
	c.	What are Green Fuels? Explain the production of hydrogen by electrolysis of water and mention its advantages.	7	L1,2	CO1
Module – 2					
Q.3	a.	Define Metallic Corrosion. Describe the electrochemical theory of corrosion taking Iron as an example.	7	L2	CO2
	b.	What is Anodizing? Explain the process of anodizing of Aluminium.	6	L2	CO2
	c.	What is CPR? A thick brass sheet of area 400 inch ² is exposed to moist air. After 2 years of period, it was found to experience a weight loss 375g due to corrosion. If the density of brass is 8.73tg/cm ³ . Calculate CPR in mpy and mmpy.	7	L2	CO2
OR					
Q.4	a.	What is meant by metal finishing? Mention the technological importance of metal finishing.	6	L1	CO2
	b.	Define electrolessplating? Explain the electroless plating of Nickel.	7	L2	CO2
	c.	Explain the electroplating of chromium as hard and decorative.	7	L2	CO3

Module – 3					
Q.5	a.	Explain the synthesis, properties and application of CPVC.	7	L3	CO3
	b.	What is composite? Explain the properties and industrial application of carbon based reinforced composites.	7	L2	CO3
	c.	Explain the synthesis, properties and industrial application of PMMA.	6	L2	CO3
OR					
Q.6	a.	What are lubricants? Explain the properties and applications of lubricants.	6	L2	CO3
	b.	Explain the synthesis, properties and application polyester.	7	L2	CO3
	c.	In a polymer, 100 molecules have molecular mass 10^3 g/mol, 250 molecules have molecular mass 10^4 g/mol, and 300 molecules have molecular mass 10^5 g/mol. Calculate the number average and weight average molecular mass of the polymer.	7	L2	CO3
Module – 4					
Q.7	a.	Define phase rule, explain the following terms with an example phase, components and degree of freedom.	7	L1	CO4
	b.	Explain the estimation of FAS potentiometrically using platinum and calomel electrode as potentiometric sensors.	7	L2	CO4
	c.	Explain the determination of pH of beverages using pH sensor glass electrode.	6	L3	CO4
OR					
Q.8	a.	Explain the lead – silver two components system along with phase diagram.	7	L2	CO2
	b.	Explain the estimation of copper present in a solution by using optical sensor method.	7	L2	CO4
	c.	Explain the determination of pH using glass electrode as pH sensing electrode.	6	L2	CO4
Module – 5					
Q.9	a.	Explain the synthesis of nonomaterial by co-precipitation method.	7	L2	CO5
	b.	Explain composition, properties and applications of stainless steel and Brass.	7	L2	CO5
	c.	Explain the following size dependent properties. i) Surface area ii) Catalytic iii) Thermal properties of nanomaterial.	6	L2	CO5
OR					
Q.10	a.	Explain the properties and application of Graphene.	7	L2	CO5
	b.	Explain the synthesis of Nano material by Sol-gel method.	6	L2	CO5
	c.	Explain the properties and applications of perovskities.	7	L2	CO5

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BCHE102/202

**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Applied Chemistry for CSE Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. M : Marks , L: Bloom's level , C: Course outcomes

Module – 1			M	L	C
Q.1	a.	Define the following terms : i) Sensor ii) Transducer iii) Actuator	6	L1	CO1
	b.	Discuss the principle, working and applications of electro chemical sensors.	7	L2	CO1
	c.	Explain the detection of ascorbic acid using disposable sensor.	7	L2	CO1
OR					
Q.2	a.	Define a battery. Give the classification of batteries with examples.	6	L1	CO1
	b.	Explain the construction and working of Li-ion battery. Mention any four applications.	7	L2	CO1
	c.	Discuss construction and working of Quantum Dot Sensitized Solar Cell. (QDSSC)	7	L2	CO1
Module – 2					
Q.3	a.	Mention any Four properties and used of QLED.	6	L1	CO2
	b.	Discuss classification of liquid crystals. Mention any four properties and applications of liquid crystals.	7	L2	CO2
	c.	Explain the types of organic memory devices by taking P-type and n-type semiconducting materials.	7	L2	CO2
OR					
Q.4	a.	Write any Four properties and applications of Polythiophenes (P ₃ HT) suitable for optoelectronic devices.	6	L1	CO2
	b.	What are memory devices? Explain the classification of electronic memory devices with suitable examples.	7	L2	CO2
	c.	Define optoelectronic device. Explain the working principle of optoelectronic device.	7	L2	CO2
Module – 3					
Q.5	a.	Define metallic corrosion. Write the steps involved in the electro chemical theory of corrosion by taking rusting of Iron as an example.	6	L1	CO3

	b.	What is CPR? A thick brass sheet of area 400 inch ² is exposed to moist air, after 2 years of period it was found to experience a weight loss 375g due to corrosion. If the density of brass is 8.73 g/cm ³ , calculate CPR in mpy and mmpy units? K = 87.6 (mmpy), K = 534(mpy)	7	L2	CO3
	c.	Explain the principle, instrumentation and working of conductometry.	7	L2	CO3
OR					
Q.6	a.	What are reference electrodes? Write the construction and working of calomel electrode with a neat labelled diagram.	6	L1	CO3
	b.	Define concentration cell? Emf of the cell Ag/Ag NO ₃ (0.01M)//Ag NO ₃ (X M)/Ag is 0.0659V at 298K. Write cell reactions and calculate the value of "X".	7	L2	CO3
	c.	Briefly explain the principle, instrumentation and working of potentiometry.	7	L2	CO3
Module – 4					
Q.7	a.	What is green fuel? Mention the advantages of green fuel (Hydrogen).	6	L1	CO4
	b.	A polymer sample contains 100 molecules of mol. Mass is 2×10^4 g/mol 300 molecules of molecular mass 3×10^3 g/mol and 500 molecules of molecular mass 5×10^3 g/mol. Calculate the number and weight average molecular weight of polymer.	7	L2	CO4
	c.	Describe the generation of hydrogen gas by alkaline water electrolysis with a neat labelled diagram.	7	L2	CO4
OR					
Q.8	a.	What is Photovoltaic cell? Write the construction and working of PV cell with a neat labelled diagram.	6	L2	CO4
	b.	Explain the preparation, properties and commercial applications of graphene oxide.	7	L2	CO4
	c.	Discuss the conduction mechanism in polyacetylene.	7	L2	CO4
Module – 5					
Q.9	a.	Define e-waste? Mention the sources and composition of e-waste.	6	L1	CO5
	b.	Briefly discuss the various steps involved in recycling of e-waste.	7	L2	CO5
	c.	Explain health hazards due to exposure of e-waste.	7	L2	CO5
OR					
Q.10	a.	Write a brief note on role of stake holders such as producers, consumers, recyclers and statutory bodies in managing the e-waste.	6	L1	CO5
	b.	Discuss the following: i) Pyrometallurgy ii) Hydrometallurgy.	7	L2	CO5
	c.	Explain the steps involved in the extraction of gold from e-waste.	7	L2	CO5

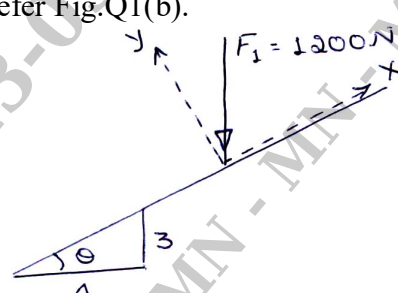
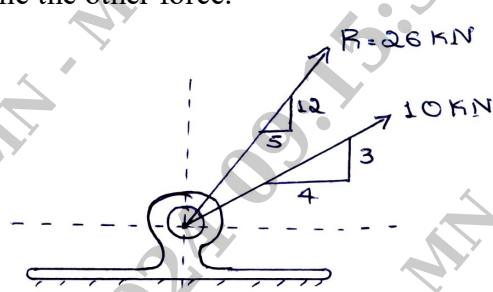
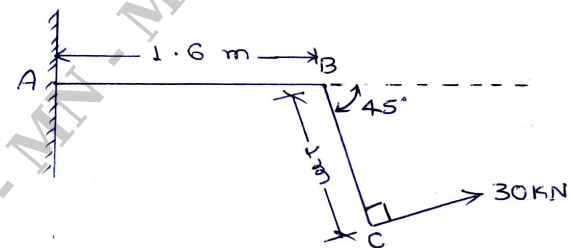
First/Second Semester B.E./B.Tech Degree Supplementary Examination, June/July 2024

Engineering Mechanics

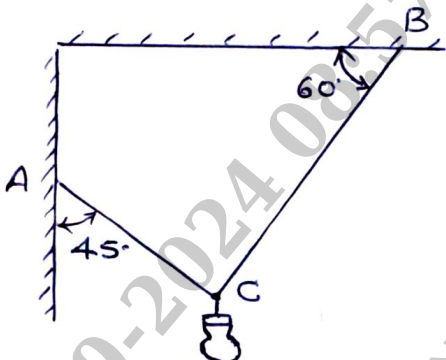
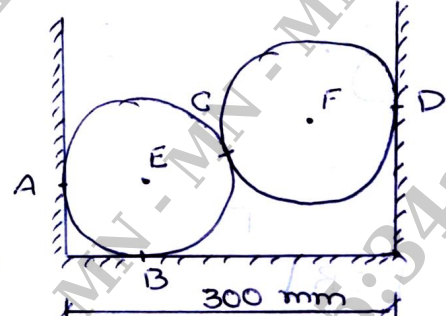
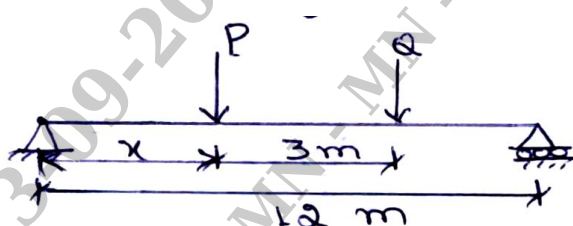
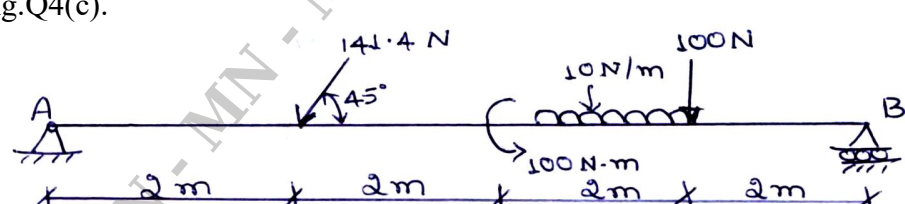
Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.
3. Missing data, if any, may be suitably assumed.*

Module – 1				M	L	C
Q.1	a.	Elaborate the concept of idealization.		6	L1	CO1
	b.	A force $F_1 = 1200\text{N}$ is acting vertically on an incline. Find its components along x and y axes. Refer Fig.Q1(b).  Fig.Q1(b)		6	L3	CO1
	c.	The 26kN force is the resultant of two forces. One of which is as shown in Fig.Q1(c). Determine the other force.  Fig.Q1(c)		8	L3	CO1
OR						
Q.2	a.	Explain briefly: i) Principle of transmissibility ii) Resolution of composition of forces		6	L2	CO1
	b.	Find moment of force about A and B for the 30kN force shown in Fig.Q2(b).  Fig.2(b)		6	L3	CO1
	c.	State and prove Varignon's theorem and its application.		8	L2	CO2

Module – 2

Q.3	a.	State and prove Lami's theorem.	6	L2	CO2
	b.	An electric bulb weighing 150N is suspended between wall and roof by two wires as shown in Fig.Q3(b). Determine the tension in the wires using Lami's theorem.  Fig.Q3(b)	6	L3	CO2
	c.	Two spheres each of radius 100mm and weight 5kN are in a rectangular box as shown in Fig.Q3(c). Calculate the reactions at all points of contact.  Fig.Q3(c)	8	L3	CO2
OR					
Q.4	a.	Explain types of loads and supports.	6	L1	CO2
	b.	Determine the distance 'x' of the load 'P' from the support A, if the reaction R_A is twice as great as reaction R_B. Take $P = 2 \text{ kN}$, $Q = 1 \text{ kN}$. Refer Fig.Q4(b).  Fig.Q4(b)	6	L3	CO2
	c.	Determine the reactions at 'A' and 'B' for the loaded beam shown in Fig.Q4(c).  Fig.Q4(c)	8	L3	CO2

Module – 3

Q.5	a.	Explain different types of Trusses.	6	L1	CO3
	b.	Find forces in members of king post truss shown in Fig.Q5(b) using method of joints and tabulate member forces.	14	L3	CO3

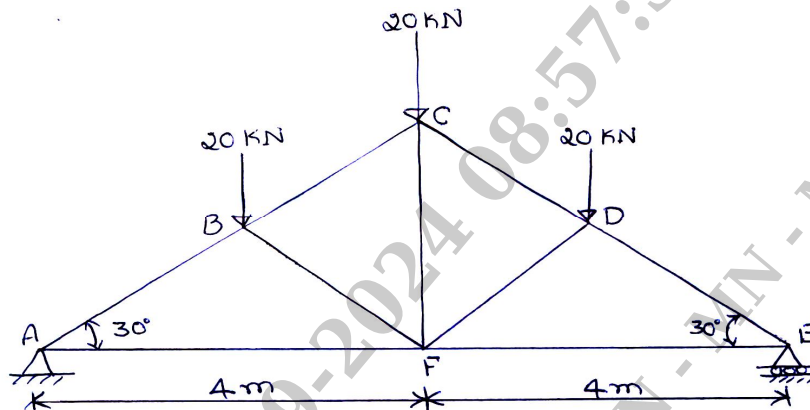


Fig.Q5(b)

OR

Q.6	a.	State Laws of Friction.	6	L1	CO3
	b.	A block weighting 4000N is resting on a horizontal surface, it supports another block of 2000N as shown in Fig.Q6(b). Find the horizontal force F just to move the block to the left. Take co-efficient of friction as 0.2.	14	L3	CO3

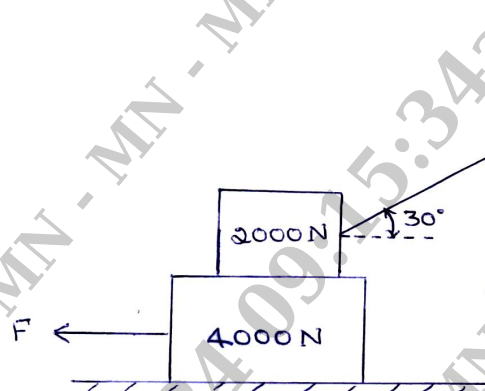


Fig.Q6(b)

Module – 4

Q.7	a.	Derive the expression for Centroid of a semicircle from first principle.	8	L3	CO4
	b.	Locate the Centroid of lamina shown in Fig.Q7(b), with respect to point 'A'.	12	L3	CO4

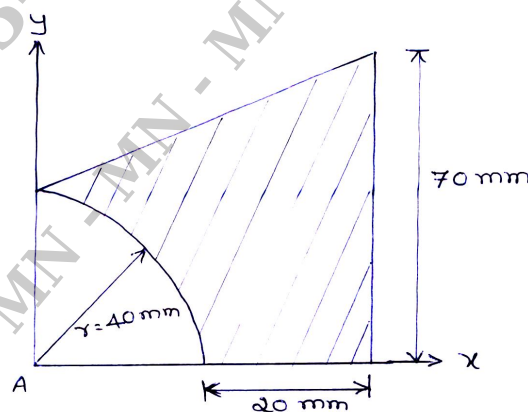
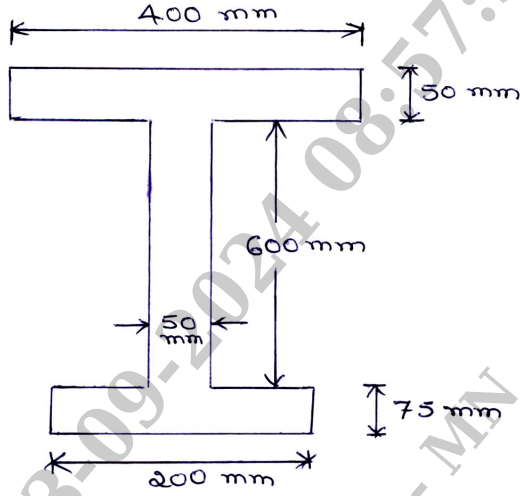


Fig.Q7(b)

OR

Q.8	a.	State and prove parallel axes theorem.	8	L3	CO4
	b.	Find the M.I. along the horizontal axis passing through the centroid of the section shown in Fig.Q8(b).  Fig.(b)	12	L3	CO4

Module – 5

Q.9	a.	A stone is thrown vertically upwards and returns to the earth in 10 secs. What was its initial velocity and how high did it go?	10	L3	CO5
	b.	Two cars P and Q accelerate from a standing start. The acceleration of 'P' is 1.3m/sec^2 and that of 'Q' is 1.6m/sec^2 . If 'Q' was originally 6m behind 'P', how long it takes to overtake 'P'?	10	L3	CO5

OR

Q.10	a.	State and explain D'Alembert's principle and its applications.	8	L2	CO5
	b.	A car travelling at a speed of 75 kmph applies brake and comes to a halt after skidding 60m. Determine : i) The deceleration ii) Time to stop the car iii) Co-efficient of friction between road and tyres.	12	L3	CO5

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BEMEM103/203

**First/Second Semester B.E./B.Tech.Degree Supplementary Examination,
June/July 2024**

Elements of Mechanical Engineering

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.
3. Use of Steam tables is permitted.*

Module – 1			M	L	C
Q.1	a.	Discuss the various emerging trends and technologies in manufacturing sector.	8	L2	CO1
	b.	Showing all the components, explain the principle of electric power generation from Hydro Power plants.	8	L2	CO1
	c.	Differentiate between Renewable and Non-Renewable Energy resources. (any four)	4	L2	CO1
OR					
Q.2	a.	With a neat sketch, explain the working principle of Thermal power plant.	10	L2	CO1
	b.	Define the following : (i) Wet steam. (ii) Super heated steam (iii) Dryness fraction. (iv) Sensible heat. (v) Latent heat	10	L1	CO1
Module – 2					
Q.3	a.	What are various methods of producing Taper turning method? Explain taper turning by swiveling the compound method.	8	L2	CO2
	b.	With a block diagram, briefly explain the basic components of CNC.	8	L2	CO2
	c.	Mention the advantages and applications of CNC. (any 2 advantages and any 2 applications)	4	L2	CO2
OR					
Q.4	a.	Explain the following with suitable sketches: (i) Upmilling (ii) Down milling (iii) Plane milling (iv) Slot milling	10	L2	CO2
	b.	Explain the following operation on lathe with suitable sketches : (i) Turning (ii) Knurling (iii) Facing (iv) Thread cutting	10	L2	CO2
Module – 3					
Q.5	a.	Explain with neat sketch, construction and working principle of 4-stroke diesel engine with the help of P-V diagram.	10	L2	CO2
	b.	With a neat sketch, explain the working principle of vapour compression refrigeration system.	10	L2	CO2

OR					
Q.6	a.	Explain the Ideal properties of Refrigerant.	6	L2	CO2
	b.	List any four applications of an Internal Combustion Engine.	4	L2	CO2
	c.	With a neat sketch, explain the working principle of Room Air Conditioner.	10	L2	CO2
Module – 4					
Q.7	a.	Briefly explain the working principle of Electric Arc Welding with a neat sketch.	10	L2	CO4
	b.	Differentiate between Belt drives and Gear drives. (any 4 points)	4	L2	CO3
	c.	A compound gear train consists of 4 gears, P, Q, R, S having 20, 40, 60 and 80 teeth respectively. The gear P is keyed to driving shaft, Q and R are compound gears, Q meshing with P and R meshes S. If P rotates at 150 rpm, what is the speed of gear S? Show gear arrangement.	6	L3	CO4
OR					
Q.8	a.	With a neat sketch, explain the working principle of Tungsten Inert Gas (TIG) welding.	10	L2	CO4
	b.	A V-belt drive transmits 10 KW power at 200 rpm. The grooved pulley has a mean diameter of 1.2 m and groove angle is 45° . Taking the co-efficient of friction as 0.25 and the angle of lap as 190° , determine the tension on each side of the belt.	10	L3	CO4
Module – 5					
Q.9	a.	With a block diagram, briefly explain the open loop and closed loop mechatronics systems.	10	L2	CO3
	b.	Briefly explain the basic components of Hybrid vehicle with block diagram.	10	L2	CO3
OR					
Q.10	a.	Explain the advantages and applications of Robots in Industries.	10	L2	CO3
	b.	Define Industrial Robot. With suitable sketches, explain various types of joints which relates to the Robot configuration.	10	L2	CO3

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CBCS SCHEME

BENGK106/206

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Question Paper Version : A

**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024
Communicative English**

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. The direction in which the formal communication flows is always _____.
a) upward b) downward c) horizontal d) All of these
 2. Communication with birds and animals is known as _____.
a) Extra personal communication b) birdistic
c) animalistic d) both b and c
 3. _____ can be defined as an unplanned interference in the communication environment causing hindrance
a) noise b) channel c) message d) direction
 4. _____ decodes the message.
a) Sender b) Receiver c) both a and b d) channel
 5. _____ is an example for downward communication
a) message b) phone call c) both a and b d) memo
 6. Different people have different perceptions about the same reality
a) True b) False c) possible d) may be
 7. Business communication is both highly formal and structural
a) True b) False c) possible d) some times
 8. Communication is a _____ process.
a) one way b) Two way c) both a and b d) Three way

9. The term communication originates from the Latin word _____ which means 'to share'.
a) communico b) communicare c) communion d) communis
10. Broadly speaking technical communication involves both formal and informal expressions in communication.
a) True b) False c) possible d) some times
11. Total number of sounds in English is _____.
a) 12 b) 08 c) 24 d) 44
12. According to phonetic method, the unit of a word is
a) word b) letter c) sound d) sentence
13. Dipthong is a _____.
a) pure vowel sound b) consonant sound
c) vowel glides or mixed vowel sounds d) both a and b
14. The full form of RP is _____.
a) Received pronunciation b) Right pronunciation
c) Reading pronunciation d) Revised pronunciation
15. The full form of IPA is _____.
a) International Phonetic Alphabet b) International Phonic Alphabet
c) International Phonetics Alphabet d) Internation Pronounced Alphabet
16. Aspiration is _____.
a) forceful release of air b) sound
c) song d) speaking
17. Which of the following has / a: / sound?
a) game b) gate c) about d) aunt
18. Which of the following has / θ / sound?
a) there b) this c) health d) father
19. Identify the word with one syllable.
a) architecture b) college c) glass d) phonetics
20. The word 'possible' has _____ syllables.
a) 4 b) 2 c) 3 d) 5
21. The structure of the word 'MAT' is _____.
a) CVC b) CCV c) CVV d) VC
22. Which of the following word has a silent letter?
a) chair b) tables c) comb d) pen
23. Which of the following word has a non-silent letter?
a) educate b) epitome c) come d) wealth

- Fill in the blanks with the correct answer :**

- Select the correct spelling :**

- Choose the correct prefix or suffix:**

- Synonyms : Choose the word that is closest in meaning:**

- 3 of 4

Antonyms : Choose the word that is opposite in meaning:

37. Barbaric :
 a) civilized b) barbarous c) cruel d) harsh
38. Deceased :
 a) healthy b) unhealthy c) alive d) dead

Choose the correct question tag :

39. You study daily, _____ ?
 a) don't you? b) will you?
 c) do you? d) wont you?
40. Pass me the salt, _____ ?
 a) could you? b) will you?
 c) dont you? d) do you?

Minimal pairs : Choose the correct pair of words from the given options :

41. Time : _____
 a) tan b) dime c) tin d) tang
42. Sheep : _____
 a) dear b) beep c) break d) brick

One word substitution : Select the appropriate word from the given options:

43. A person who believes in God.
 a) Theist b) atheist c) atheism d) sadist
44. One who understands the finer aspects of art, music, etc.
 a) connoisseur b) epicurian c) cosmopolitan d) pilgrim
45. The full form of MTI is _____
 a) Most Talented Inventor b) Mother Tongue Influence
 c) Mother Tongue Intelligence d) Most Talked Influence
46. _____ is an important element of body language.
 a) Personal appearance b) Speaking
 c) Crossed arms d) Blank expression
47. Voice modulation is essential for an effective speech.
 a) True b) False c) may be d) possible
48. Non-verbal communication effectively reinforces verbal communication.
 a) True b) False c) possible d) may be
49. A speaker who speaks too loudly at times puts away the listeners in the front rows.
 a) True b) False c) might be d) not always
50. In public speaking situations the rate/pace of our delivery should be rapid fire and urgent.
 a) True b) False c) may be d) always

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CBCS SCHEME

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BESCK104A/BESCKA104

**First Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

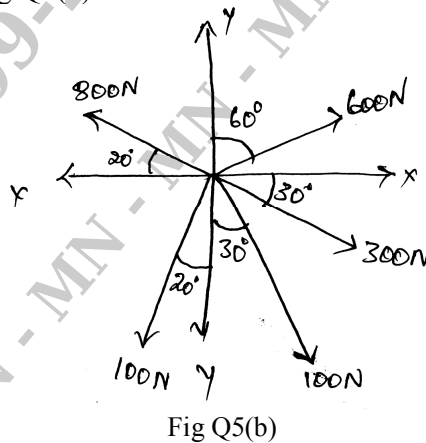
Introduction to Civil Engineering

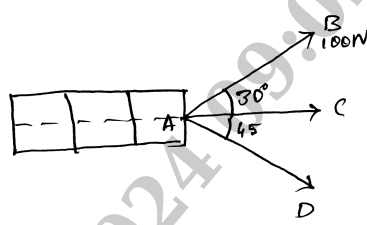
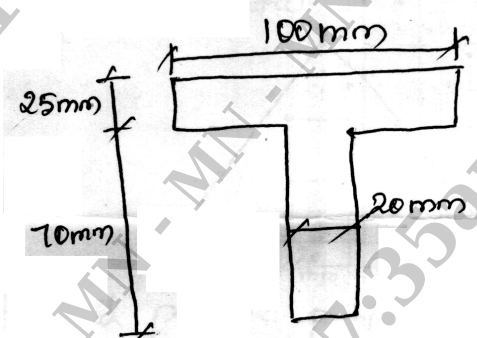
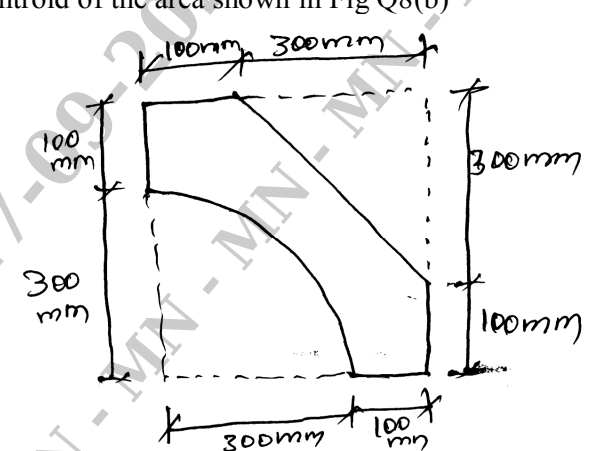
Time: 3 hrs.

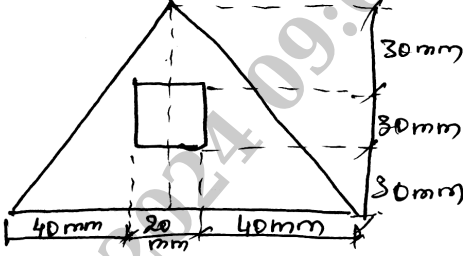
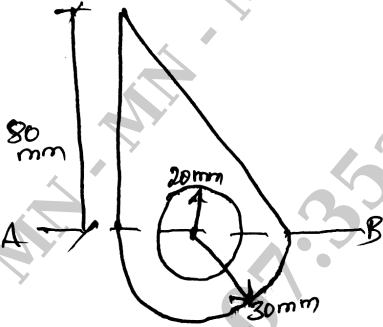
Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the scope of following branches of Civil Engineering : i) Surveying ii) Environmental Engineering.	10	L1	CO1
	b.	Explain the structural elements i) Plinth ii) Chejja iii) Staircases.	10	L1	CO1
OR					
Q.2	a.	Explain the scope of : i) Hydraulic and Water Resource Engineering ii) Transportation engineering	10	L1	CO2
	b.	Explain any five chemicals used in construction.	10	L1	CO2
Module – 2					
Q.3	a.	Explain the development made towards the sustainability of the earth.	10	L1	CO2
	b.	Explain the concept of recycling of waste materials.	10	L1	CO2
OR					
Q.4	a.	Differentiate between safe city and smart city.	10	L1	CO2
	b.	Explain different methods to control urban flood.	10	L2	CO2
Module – 3					
Q.5	a.	Explain the basic idealization of engineering mechanics.	5	L2	CO3
	b.	Find the magnitude and direction of the resultant of the coplanar force system shown in Fig Q5(b)	15	L3	CO3



OR					
Q.6	a.	State is explain principle of superposition and principle of transmissibility.	10	L2	CO2
	b.	Fig Q6(b) show that the top view of car pulled by two cables AB and AD. The car is moving along AC. If the force in cable AB is 100N. Calculate the force in AD and the resultant.	10	L3	CO3
 Fig Q6(b)					
Module – 4					
Q.7	a.	Derive an expression for determining centroid of a rectangle using first principle.	10	L2	CO4
	b.	Derive the centroid of the lamina shown in Fig Q7(b)	10	L3	CO4
 Fig Q7(b)					
OR					
Q.8	a.	Derive an expression for determining centroid of a triangle using first principle.	10	L2	CO4
	b.	Derive the centroid of the area shown in Fig Q8(b)	10	L3	CO4
 Fig Q8(b)					

Module – 5					
Q.9	a.	Explain moment of inertia, polar moment of inertia and radius of gyration with sketches.	10	L2	CO5
	b.	Determine the moment of inertia about centroidal horizontal X-X axis shown in Fig Q9(b)	10	L3	CO5
 Fig Q9(b)					
OR					
Q.10	a.	State and prove perpendicular axis theorem.	10	L2	CO5
	b.	Determine the moment of inertia about the given axis AB as shown in Fig Q10(b)	10	L3	CO5
 Fig Q10(b)					

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BESCK104B / BESCKB104

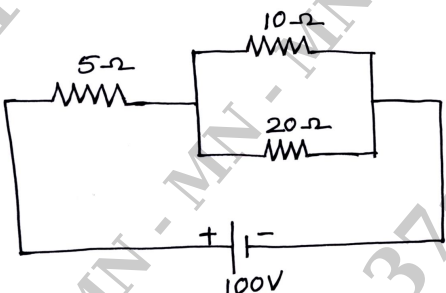
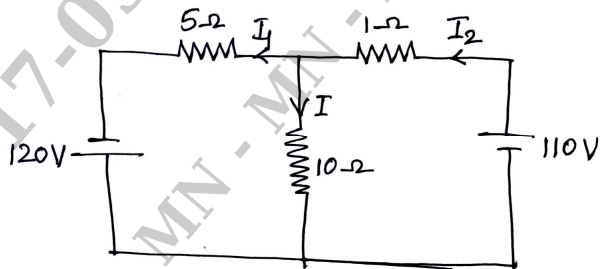
First Semester B.E./B.Tech. Degree Supplementary Examination, June/July 2024

Introduction to Electrical Engineering

Time: 3 hrs.

Max. Marks: 100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. VTU Formula Hand Book is permitted.
 3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Explain the general structure of electrical power system using single line diagram approach.	06	L2	CO5
	b.	Find the current and power in each resistor for the circuit shown in Fig.Q1(b).  Fig.Q1(b)	06	L3	CO2
	c.	Explain the operation of Nuclear power generation plant with the help of block diagram.	08	L2	CO1
OR					
Q.2	a.	State and explain Kirchoff's laws.	06	L2	CO1
	b.	Distinguish between conventional and nonconventional sources of energy.	06	L3	CO1
	c.	Find the current I_1 , I_2 and I in the network shown in Fig.Q2(c).  Fig.Q2(c)	08	L3	CO2
Module – 2					
Q.3	a.	Define the following terms: (i) Average value (ii) RMS value (iii) Phase difference (iv) Amplitude	06	L1	CO2

BESCK104B / BESCKB104

	b.	A pure inductor excited by sinusoidal varying AC voltage, show that the average power consumed by inductor is zero. Also draw the voltage, current and power waveforms.	08	L2	CO2
	c.	A star connected load consists of 6Ω resistance and 8Ω inductive reactance in each phase. A supply of 440 V at 50 Hz is applied to the load. Find the line current, power factor and power consumed by the load.	06	L3	CO2
OR					
Q.4	a.	Develop an equation for the power consumed by an RC series circuit. Draw the waveforms of voltage, current and power.	08	L2	CO2
	b.	What are the limitations and advantages of three phase system?	06	L3	CO2
	c.	A circuit consists of resistance 10Ω , an inductance of 16 mH and a capacitance of $150\ \mu\text{F}$ connected in series. A supply of 100 V at 50 Hz is given to the circuit. Find the current, power factor and power consumed.	06	L3	CO2
Module – 3					
Q.5	a.	With the help of neat diagram, explain the construction of D.C generator.	08	L2	CO3
	b.	Give the classification of DC generator. Obtain the expression for EMF equation of a DC generator.	06	L2	CO4
	c.	A four pole d.c. shunt motor takes 22.5 Amps from a 250 V supply, $R_a = 0.5\ \Omega$ and $R_{sh} = 125\ \Omega$. The armature is wave connected with 300 conductors. If the flux per pole is 0.02 wb, calculate: (i) The speed (ii) Torque developed (iii) Power developed	06	L3	CO4
OR					
Q.6	a.	What is back emf in a dc motor? What is its significance?	06	L1	CO4
	b.	Sketch N-I and T-I characteristics of DC: (i) Series (ii) Shunt motors. Mention two applications of each motor.	08	L4	CO4
	c.	A 8 pole wave wound DC shunt generator has 36 slots, 10 conductors in each slot. The flux/pole is 0.01 wb. It runs at 1200 rpm. The armature and field resistance are $0.1\ \Omega$ and $100\ \Omega$ respectively. Calculate the terminal voltage when the load current is 120 A. Neglect armature reactions.	06	L3	CO4
Module – 4					
Q.7	a.	Discuss various types of losses in a transformer.	08	L2	CO4
	b.	With neat sketch, explain the construction and working principle of 1 ϕ transformer.	06	L2	CO3
	c.	A 3 phase induction motor with 4 poles is supplied from an alternator having six poles and running at 1000 rpm. Calculate: (i) Synchronous speed of the IM (ii) Its speed when slip is 0.04 (iii) Frequency of the rotor emf when the speed is 600 rpm	06	L3	CO4

OR

Q.8	a.	With relevant diagram, explain the construction of three phase induction motor.	08	L2	CO3
	b.	Define slip of a 3-phase induction motor and derive the relation between supply frequency and rotor current frequency.	06	L2	CO4
	c.	The primary winding of a 25 KVA transformer has 200 turns and is connected to 230 volts, 50 Hz supply. The secondary turns are 50. Calculate: (i) No load secondary induced emf (ii) Full load primary and secondary currents (iii) The flux density in the core, if the cross section of the core is 60 cm ² .	06	L3	CO4

Module – 5

Q.9	a.	Mention the different types of wiring with relevant circuit diagram and switching tables, explain two-way and three way control of lamps.	08	L2	CO5
	b.	Define tariff. Explain briefly the two part tariff with its advantages and disadvantages.	06	L2	CO5
	c.	Explain the working principle of fuse and MCB.	06	L2	CO5

OR

Q.10	a.	What is earthing? Why earthing is required? With the help of neat sketch, explain plate earthing.	08	L2	CO5
	b.	Write a short note on precautions against an electric shock.	06	L2	CO5
	c.	List out the power rating of household appliances including air conditioners, PCs, Laptops, printers etc. Find the total power consumed.	06	L2	CO5

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First Semester B.E/B.Tech. Degree Supplementary Examination, June/July 2024

Introduction to Electronics and Communication

Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
1	a.	Describe full wave bridge rectifier with relevant waveform? Memorize its measurable parameters.	7	L2	CO1
	b.	Explain how zener diode acts as voltage regulator. Discuss line and load regulation.	7	L2	CO1
	c.	Illustrate the difference between zener and avalanche breakdown.	6	L2	CO1
OR					
2	a.	Discuss half wave rectifier with capacitor filter. Memorize ripple factor.	7	L2	CO1
	b.	Describe with neat sketch and working operation of RC coupled amplifier and specify frequency response with parameters.	7	L2	CO1
	c.	Discriminate half wave rectifier and full wave rectifier.	6	L2	CO1
Module – 2					
3	a.	Describe the operation of crystal oscillator and state its applications.	7	L2	CO2
	b.	Discuss any two applications of OPAMP.	7	L2	CO2
	c.	State ideal and practical characteristics of OPAMP.	6	L2	CO2
OR					
4	a.	Explain the concept for sustained oscillations of ladder type network oscillator.	7	L2	CO2
	b.	Define multivibrator and also discuss single stage astable oscillator using OPAMP.	7	L2	CO2
	c.	Describe various blocks of OPAM and discuss its measurable parameter.	6	L2	CO2
Module – 3					
5	a.	i) Convert $(725.25)_8$ to $()_{10}$ and $()_2$ ii) Determine the value of x if $(211)_x = (152)_8$	7	L3	CO3
	b.	Discuss the different theorems and postulates of Boolean Algebra and prove each of them with truth table.	7	L2	CO3
	c.	Find the complement of functions F_1 and F_2 i) $F_1(x, y, z) = x'yz' + x'y'z$ ii) $F_2(x, y, z) = x(y'z' + yz)$.	6	L3	CO3

OR

6	a.	Simplify and realize the expression using NAND and NOR $F = AB + AC + BD + CD$.	7	L3	CO3
	b.	Implement full adder circuit with truth table and deduce expression for sum and carry.	7	L2	CO3
	c.	Implement EX–OR and EX–NOR using NAND.	6	L2	CO3

Module – 4

7	a.	Define general computing system. Describe its classification in contrast to embedded system.	7	L2	CO4
	b.	Compare microprocessor and microcontroller with their features.	7	L2	CO4
	c.	Explain the following : i) Transducers ii) Actuators.	6	L2	CO4

OR

8	a.	Explain various elements of embedded systems and state its applications.	7	L2	CO4
	b.	Distinguish RISC and CISC.	6	L2	CO4
	c.	Explain : i) 7 segment LED display ii) Sensor.	7	L2	CO4

Module – 5

9	a.	Explain the concept of AM wave and interpret MI and transmission efficiency.	7	L2	CO5
	b.	Discuss the various modes of radio wave propagation.	7	L2	CO5
	c.	Describe ASK used in communication system.	6	L2	CO5

OR

10	a.	Discuss frequency modulation in communication system and describe MI and frequency deviation.	7	L2	CO5
	b.	Describe the various block of communication system.	6	L2	CO5
	c.	Explain PSK modulation technique in communication system.	7	L2	CO5

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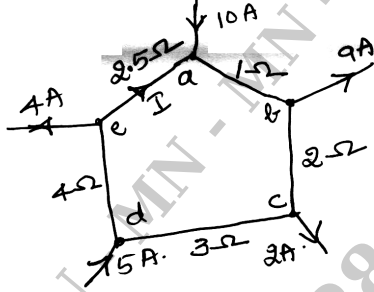
Second Semester B.E./B.Tech. Degree Supplementary Examination, June/July 2024

Introduction to Electrical Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	With a neat single line diagram, explain various components of electrical power transmission and distribution system.		6	L2	CO1
	b.	State and explain Ohm's law with its limitations.		6	L2	CO1
	c.	Find the currents in all the branches of the network given below starting from branch current 'I' in branch 'ea' Ref Fig Q1(c)		8	L3	CO1
 Fig Q1(c)						
OR						
Q.2	a.	With a neat block diagram, explain a Nuclear power plant.		6	L2	CO1
	b.	State and explain Kirchhoff's laws with relevant examples for each.		6	L2	CO1
	c.	A resistor of 12Ω is connected in series with a combination of 15Ω and 20Ω resistors in parallel. The supply voltage is 120V. Find the current flowing through the parallel branches and the circuit. Also find voltage across each of them.		8	L3	CO1
Module – 2						
Q.3	a.	With the help of a circuit diagram and the phasor diagram, conclude about phase angle, impedance and power consumed by a series R-C circuit.		6	L2	CO2
	b.	If the details given are for a 3φ, 50Hz, 400V supply where $Z_{ph} = 25 + j22\Omega$ are connected in delta. Calculate the line current, phase current, power factor, active power, reactive power and apparent power.		8	L3	CO2
	c.	A series circuit consisting of R, L, C with respective voltage of 170V, 150V and 100V across them. The current flowing through them is 4A. Find the values of R, L, C, power factor and also the supply voltage. Assume $f = 50\text{Hz}$.		6	L3	CO2

OR				
Q.4	a.	For single phase sinusoid generated emf, define : i) rms value ii) average value iii) form factor iv) peak factor v) phase vi) phase difference.	6	L1 CO2
	b.	Show that the current through a purely inductive circuit lags the applied voltage by 90° . Also discuss about the average power consumed. Draw the wave shapes of current, voltage and power.	8	L2 CO2
	c.	A circuit consists of a resistance of 10Ω and an inductance of 16mH and a capacitor of $150\mu\text{F}$ in series across a supply of 240V , 50Hz . Find : i) current ii) power factor iii) power consumed. Draw the phasor diagram.	6	L3 CO2
Module – 3				
Q.5	a.	With a neat diagram, explain the construction of a DC machine.	8	L2 CO3
	b.	Derive an expression for armature torque developed in a DC motor.	6	L2 CO3
	c.	An 8 pole lap wound DC generator has 500 armature conductors and has a useful flux per pole of 0.065Wb . What is the generated emf, if it runs at 1000rpm ? When the armature is wave wound for the same, what will be its speed to generate the same emf.	6	L3 CO3
OR				
Q.6	a.	Explain various methods of speed control of DC shunt motor.	6	L2 CO3
	b.	With usual notations, derive emf equation of a DC generator.	8	L2 CO3
	c.	A 6 pole, lap connected DC shunt motor has 492 armature conductors having resistance of 0.2Ω . The flux per pole available is 50mWb . The motor runs at 20rps across a supply of 500V at full load. What will be the speed of motor when the load is reduced by 50%. Neglect all drops connected to brush and armature reaction.	6	L3 CO3
Module – 4				
Q.7	a.	Explain various losses incurred in a single phase transformer. Also discuss about technique used to minimize them.	6	L2 CO3
	b.	With neat vectorial representation, explain rmf theory.	8	L2 CO3
	c.	The primary winding of a 25KVA , single phase transformer has 200 turns and is connected to a 230V , 50Hz supply. The secondary number of turns is 50. Calculate : i) No load secondary emf ii) Primary and secondary currents iii) Flux density in the core, if the area of cross section is 60cm^2 .	6	L2 CO3
OR				
Q.8	a.	With neat schematic diagrams, compare core and shell type of transformer.	6	L2 CO4
	b.	With diagrams, explain the types of three phase induction motors.	8	L2 CO4

	c.	A 12 pole, 3 phase alternator runs at 500rpm is coupled to a 3 phase induction motor which runs at full load speed of 1440rpm. Comment on synchronous speed of the induction motor. Calculate percentage slip and number of poles of the induction motor.	6	L3	CO4
Module – 5					
Q.9	a.	With neat circuit diagrams and switching tables, explain two way and three way control of a load.	8	L2	CO5
	b.	Explain plate earthing with a neat diagram.	6	L2	CO5
	c.	Explain working of a fuse. Also explain rated current, fusing current and fusing factor.	6	L2	CO5
OR					
Q.10	a.	What is electric shock? List preventive measure against the shock.	8	L2	CO5
	b.	Explain “unit” used in electrical energy consumption. Also explain two part tariff with its advantages and disadvantages.	6	L2	CO5
	c.	Calculate the monthly billing of a domestic consumer whose load factor is 70% if a unit is charges at Rs. 3. The load detail are as below : 5 numbers of 10W LED lamps 2 numbers of 80W fans 1 number of 450W AC 1 number of 2kW Geyser 1 number of 800W Refrigerators.	6	L3	CO5

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BESCKE204/BESCK204E

**Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Introduction to C Programming

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain basic organization of a computer.	7	L2	CO1
	b.	Explain the terms Computer and its characteristics.	7	L2	CO1
	c.	Explain classification of Computers.	6	L2	CO1
OR					
Q.2	a.	Write a C program to take input the student data and display it.	7	L2	CO2
	b.	Write and explain the structure of C program.	7	L2	CO2
	c.	Explain Input and Output statements in C language.	6	L2	CO2
Module – 2					
Q.3	a.	Write a C program to find the largest number out of given three numbers.	7	L2	CO2
	b.	Explain type conversion in C language with example.	7	L2	CO2
	c.	Explain different types of operators in C language with the help of examples.	6	L2	CO2
OR					
Q.4	a.	Explain 'If – else' conditional branching statements in C language.	7	L2	CO2
	b.	Explain 'for' iterative statement with example.	7	L2	CO2
	c.	Explain 'goto' statement with example.	6	L2	CO2
Module – 3					
Q.5	a.	Explain following terms with examples : i) Function declaration ii) Function definition iii) Function call.	7	L2	CO4
	b.	What is Function in C program? Explain needs of function.	7	L2	CO4
	c.	Explain different types of scope of a variable with example.	6	L2	CO4
OR					
Q.6	a.	Write a C program to perform addition of array elements.	7	L2	CO3
	b.	Explain accessing of array elements with examples.	7	L2	CO3

	c.	Explain how to pass array to function with example.	6	L2	CO4
Module – 4					
Q.7	a.	Write a C program to perform string operations using library functions.	7	L2	CO4
	b.	Explain functions for reading string data from keyboard with help of example.	7	L2	CO4
	c.	Explain bubble sort technique with example.	6	L2	CO3
OR					
Q.8	a.	Write a C program to perform addition of two matrices.	7	L2	CO3
	b.	Illustrate to access elements of two dimensional arrays. Explain with example.	7	L2	CO3
	c.	Illustrate to pass two dimensional arrays to function. Explain with example.	6	L2	CO3
Module – 5					
Q.9	a.	Explain how to access members of structure data type with example.	7	L2	CO3
	b.	Explain the structure data type and declaration of structure data type and its memory allocation with example.	7	L2	CO3
	c.	Explain String taxonomy with help of example.	6	L2	CO3
OR					
Q.10	a.	What is Pointer? Explain declaration of pointer variable with example.	7	L2	CO3
	b.	Explain Arithmetic operations on pointer variable with examples.	7	L2	CO3
	c.	Explain NULL pointer and generic pointer with example.	6	L2	CO3

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BETCK205B/ BETCKB205

Second Semester B.E./B.Tech Degree Supplementary Examination, June/July 2024 **Green Buildings**

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain briefly the manufacturing process of burnt bricks.	10	L2	CO1
	b.	List the advantages of using following in construction : i) Stabilized mud block ii) Bricks.	10	L2	CO1
OR					
Q.2	a.	Explain the environmental issues related to quarrying of building materials.	10	L2	CO1
	b.	List the advantages of using following materials in construction. i) Concrete blocks ii) Bamboo.	10	L2	CO1
Module – 2					
Q.3	a.	Explain briefly any two types of precast members used in building construction.	10	L2	CO2
	b.	Explain the following in construction of building. i) Cavity wall ii) Ferro cement.	10	L2	CO2
OR					
Q.4	a.	What are the objectives of Nirmithi Kendra and COSTFORD? Explain their contributions to the society.	10	L2	CO2
	b.	Explain the following in construction of building : i) Filler slab technology ii) Rat –trap – bond.	10	L2	CO2
Module – 3					
Q.5	a.	Explain the causes and effects of global warming.	10	L2	CO3
	b.	Define carbon footprint. What are the global efforts to reduce carbon emission?	10	L2	CO3
OR					
Q.6	a.	Explain the following : i) Embodied energy in materials ii) Comparison between conventional building and green building.	10	L2	CO3
	b.	Describe the environmental, economical, health and social benefits of green building.	10	L2	CO3
1 of 2					

Module – 4

Q.7	a.	Explain the purpose and key highlights of following green rating systems : i) LEED ii) GRIHA.	10	L2	CO4
	b.	Write a note on : i) Characteristics of sustainable buildings ii) Green design.	10	L2	CO4

OR

Q.8	a.	Explain the purpose and key highlights of following green rating systems. i) BREEAM ii) GREEN STAR.	10	L2	CO4
	b.	Write short notes on : i) Principles of sustainable development ii) Sustainably managed materials.	10	L2	CO4

Module – 5

Q.9	a.	Describe the concepts of solar passive cooling and heating of building.	10	L3	CO5
	b.	Write a note on : i) Urban environment and green building ii) Zero energy building.	10	L2	CO5

OR

Q.10	a.	Explain with help of one case study on solar passive cooled and heating building.	10	L3	CO5
	b.	Explain the management of following types of waste : i) Solid waste ii) Sewage waste.	10	L2	CO5

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Question Paper Version : A

**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Indian Constitution

Time: 1 hr.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

1. Ultimate source of authority under the Constitution lies in ____
a) Parliament
b) People of India
c) Judiciary
d) Executive
2. The Constitution of India was adopted by the Constituent Assembly in the year or date ____
a) 26 - 01-1950
b) 29 - 11 - 1949
c) 15 - 08 - 1947
d) 09 - 09 - 1946
3. The basic objectives of Indian Constitution are found in ____
a) Fundamental Rights
b) Directive Principles of State Policy
c) Preamble
d) Fundamental Duties
4. Who was the Permanent Chairman of the Constituent Assembly?
a) Mahatma Gandhi
b) Jawaharlal Nehru
c) Dr. B.R. Ambedkar
d) Dr. Rajendra Prasad
5. Who was the leader of the Extremist group to fight against British Rule?
a) Sardar Vallabhai Patel
b) Mahatma Gandhi
c) Subhash Chandra Bose
d) Dr. Rajendra Prasad
6. Government of India set of 1919 introduced a system of ____
a) Federal
b) Diarchy
c) Democracy
d) Anarchy
7. The Cabinet mission came to India in the year ____
a) 1935
b) 1944
c) 1945
d) 1946
8. How much time was taken by drafting Committee to draft the Indian Constitution?
a) 2 years 11 months 18 days
b) 3 years 11 months 8 days
c) 2 years 8 months 11 days
d) 1 year 11 months 18 days

9. How many members were appointed for the drafting Committee of the Constitution excluding Chairman?
a) 5 b) 6 c) 8 d) 7
10. The term 'Unitary' means :
a) Concentration of Powers b) Division of Powers
c) Absolute Power d) All of these
11. Which is the key to open the minds of the makers of the Constitution?
a) Fundamental Rights b) Directive Principles
c) Fundamental Duties d) Preamble
12. "Rights are not only privileges but they are the weapons in the hands of people to control Government", who gave this statement?
a) Dr. Rajendra Prasad b) Dr. B.N. Rao
c) Dr. Ambedkar d) S.V. Patel
13. The practice of untouchability is prohibited under ____
a) Article 14 b) Article 15 c) Article 16 d) Article 17
14. Right to Life under Art. 21 does not include
a) Right to Education b) Right to Dignity
c) Right to Suicide d) Right to Livelihood
15. Reasonable restrictions can be imposed by the State under :
a) Art. 19 (1) (a) to (g) b) Art. 19 (2) to (6)
c) Art. 20 (a) to (c) d) Art. 21
16. Fundamental Rights under Part – III can be protected by
a) Supreme Court b) Parliament
c) Union Chief Executive d) Attorney – General of India
17. 'Writ of Mandamers' cannot be issued against :
a) State Government b) President of India
c) Prime Minister of India d) Public Servant
18. An arrested person is to be produced before the Magistrate or Court within :
a) One month b) 3 month c) 24 hours d) 48 hours
19. No Person is to be punished twice for the same offence "is The Fundamental Right available under Article :
a) 19 b) 20 c) 21 d) 22
20. What is the meaning of "Habeas Corpus"?
a) You may have the body b) To do a duty
c) On what Authority? d) None of these
21. "Right to Press" (news media) is included under :
a) Right to Freedom of Speech and expression
b) Right to Assembly
c) Right to trade d) Right to move

22. Right to Minorities are guaranteed under Articles :
 a) 14 to 16 b) 21 to 24 c) Arts. 29 & 30 d) 17 and 18
23. The Directive principles of State Policy are :
 a) Enforceable by Court b) Not enforceable by Court
 c) Only a direction to State Government d) None of these
24. Which provision of the Constitution recognizes International Law?
 a) Art. 39 b) Art. 48 c) Art. 51 d) Art. 44
25. DPSP (Part – IV) directs the State to secure to all workers :
 a) Minimum wages b) Living wages c) Standard wages d) Fair wages
26. Who is having the duty to send the children to School?
 a) State b) Parent / Guardian c) Zilla Panchayat d) None of these
27. Which Article enumerates Fundamental duties?
 a) Art. 41 b) Art. 51 c) 51 - A d) Art. 52
28. How many members are nominated to Rajya Sabha by the President?
 a) Two b) One c) Twelve d) Eight
29. Who can certify the money bill immediately after introducing in Lok Sabha?
 a) Prime Minister b) Speaker c) Finance Minister d) Any one
30. Who can dissolve Lok Sabha?
 a) Prime Minister b) Speaker c) President d) Vice - President
31. Who can pass an Ordinance in the State Government?
 a) Governor b) Chief Minister c) Law Minister d) Speaker
32. Who can appoint the Chief Justice of Supreme Court?
 a) Prime Minister b) Law Minister c) Vice - President d) President
33. What is the minimum age to become the member of Legislative Assembly?
 a) 21 years b) 25 years c) 30 years d) 35 years
34. Special provisions are given for
 a) Backward classes b) Women & Children
 c) Senior Citizens d) Both 'a' and 'b'
35. How many kinds of emergencies are incorporated in the Constitution?
 a) 5 Types b) 4 Types c) 3 Types d) 2 Types
36. Who is to act on the advice of the Council of Ministers?
 a) Prime Minister b) President c) Vice - President d) Governor
37. The President cannot issue the proclamation of emergency after 1978 for the reason :
 a) Internal disturbance b) Terrorism inside India
 c) Armed rebellion d) Covid like Epidemic diseases
38. Who can recommend for the declaration of State emergency?
 a) Chief Minister b) Governor c) State Cabinet d) None of these

39. Which are the Articles not to be suspended during National Emergency?
a) Arts. 14 to 16 b) Arts. 19 and 22 c) Arts. 20 and 21 d) Arts. 29 to 30
40. Who appoints the Chief Election Commissioner of India?
a) President b) Prime Minister
c) Chief Justice of India d) Cabinet
41. How many members are there in the Election Commission at present?
a) Seven b) Five c) Four d) Three
42. There is no provision in the Indian Constitution for the impeachment of :
a) President b) Chief Justice of Supreme Court
c) Chief Election Commissioner d) Governor
43. Election Commission does not conduct Election to
a) President b) Vice - President c) Speaker d) MPS
44. Who can amend the provisions of the Constitution?
a) Parliament b) Cabinet c) President d) State Legislative
45. The Amendment procedure laid down in the Indian Constitution under the Article :
a) Art. 324 b) Art. 360 c) Art. 368 d) Art. 378
46. Which one of the following Amendments decreased the age of voting from 21 years to 18 years.
a) 44th Amendment b) 61st Amendment c) 62nd Amendment d) 72nd Amendment
47. By which Amendment the Fundamental Duties of Citizens was included in the Constitution of India?
a) 44th b) 38th c) 41st d) 42nd
48. Who has been made responsible for free and fair Elections in the Country?
a) President b) Chief Justice of Supreme Court
c) Prime Minister d) Chief Election Commissioner
49. The duration of State emergency in the first instance is
a) One month b) Two months c) 3 months d) Six months
50. The concept of Election is based on ____
a) Federalism b) Secularism c) Socialism d) Democracy

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Question Paper Version : A

**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Innovation and Design Thinking

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. One of the following steps is the correct sequence of phase of the design thinking process.
a) Ideate, Prototype, Define, Empathize and test
b) Empathize, Define, Ideate, Prototype and test
c) Test, Prototype, Define, Ideate and Empathize
d) None of these
 2. The following is the key element of the design thinking process,
a) Only human centered, creative and playful.
b) Only Iterative and collaborative
c) Only Prototype driven
d) All of these
 3. Design thinking methodology involves one of the following distinct stages,
a) Discover and Define
b) Define and Develop
c) Develop and Deliver
d) All of these
 4. Horward Business School Dean Mr. Shrikanth Datar leverage _____ stage frame work for design thinking.
a) One stage b) Two stage c) Four stage d) Three stage
 5. The elements of design thinking are shape, colour, space, form line, value and texture.
a) True b) False

6. To launch a successful product, the team must follow _____ D
a) 4 b) 10 c) 5 d) 8
7. How many mind sets are required for design thinking?
a) 4 b) 6 c) 2 d) 8
8. _____ is the founder of design thinking.
a) Sir, Isaac Newton b) Mr. Herbert A Siman
c) Varignon d) Lami
9. The following are the 3 lenses of design thinking,
a) Resolution, Composition and Equilibrium
b) Force, Moment and Couple
c) Desirability, Viability and Feasibility
d) Desirability, acceptability and Rejection
10. All the visual designs are composed of _____ elements and these elements are combined and arranged to create a desired visual appearance.
a) 6 b) 3 c) 8 d) 4
11. The main principles of graphic design is one of the followings :
a) Balance, Contrast and Emphasize
b) Repetition and pattern and proportion
c) Movement white space, unity and variety
d) All of these
12. One of the followings is the basic design rules :
a) Contrast only b) Repetition and alignment only
c) Alignment and Proximity d) All of these
13. One of the followings is the benefit of design thinking :
a) Ability to develop product innovation b) Add value to customer's lives
c) Drives revenues to the firm d) All of these
14. One of the followings is the 3PS of design thinking,
a) Production, Transportation and Marketing
b) Immersion, Ideation and Prototyping
c) All of these
d) only (a)
15. The SWOT analysis in design thinking is one of the followings,
a) Silence, Wealth, Origin and Treat
b) Strength, Weakness, Opportunities and Threats
c) All of these
d) None of these
16. The types of collaboration is one of the followings :
a) Alliances b) Innovation c) Ecosystem d) All of these

17. The following any one is the essential elements of collaboration,
 - a) Co-operation and assertiveness
 - b) Autonomy, Responsibility and Accountability
 - c) Communication, Co-ordination, Natural trust and Respect
 - d) All of these
18. The one of the followings is the key of collaboration,
 - a) Dependable of people
 - b) Sharing vision
 - c) Voluntary
 - d) All of these
19. One of the below is not a type of collaboration working,
 - a) Mass collaboration
 - b) Team collaboration
 - c) Community collaboration
 - d) Network and Cloud collaboration
20. One of the followings is the principle of collaboration,
 - a) Trust and Empathy
 - b) Positivity and Clarity
 - c) Accountability
 - d) All of these
21. One of the following is the best principle of collaboration,
 - a) Only Relationship management
 - b) Only alignment on vision and values
 - c) Both collaborative leadership, Governance and Processes
 - d) All of these
22. One of the followings is the 4 steps of collaboration,
 - a) Forming, Norming, Storming and Performing
 - b) Fighting, Negligence, Strengthening and Personalizing
 - c) All of the these
 - d) None of these
23. One of the followings will not be included to build collaboration.
 - a) Pinpoint and Promote a purpose of collaboration
 - b) Lead by example and celebrate diverse personalities
 - c) To break the collaboration whenever required
 - d) Offer rewards or incentives for collaboration
24. One of the following is not the best example of collaboration
 - a) Brain storming
 - b) Group discussions
 - c) Prolonged time consumption
 - d) Reaching a consensus about processes or analyzing problems and finding a solution
25. The characteristics of collaboration are,
 - a) Strong leadership, clearly defined roles for subgroups
 - b) Effective and frequent communication
 - c) Periodic and temporary suppression of the ego
 - d) All of these
26. One of the following words will not describe collaboration :
 - a) Partnership and association
 - b) Unilateral, Sole
 - c) Co-operation, affiliation, connection
 - d) Relationship, Interaction and Alliances

27. The following is the way to develop collaboration,
 - a) Communicate clearly and participate in team building activities
 - b) Choose the right digital tool for the right task
 - c) Go beyond your comfort zone and become a mentor
 - d) All of these
28. One of the followings is the skill to improve collaboration,
 - a) Communication, Respect diversity
 - b) Open mindedness, knowledge sharing and debate
 - c) Being organized and delegating tasks
 - d) All of these
29. One of the followings is the advantages of collaboration,
 - a) It brings people close together
 - b) It opens up new channels for communication
 - c) It boosts across a person's organization
 - d) All of these
30. BPM in design thinking means,
 - a) Best Practices in Modeling
 - b) Business Process Modeling
 - c) All of the these
 - d) None of these
31. The phases of business process modeling are,
 - a) Information gathering, Work flow modeling
 - b) Implementation, Verification and Execution
 - c) All of these
 - d) None of these
32. The most important elements of design thinking are,
 - a) Empathy, Expansive thinking and Experimentation
 - b) Execution of work not in time and exemption from work
 - c) All of these
 - d) None of these
33. The types of product prototypes are,
 - a) Feasibility prototypes
 - b) Low fidelity user prototype
 - c) High fidelity user prototypes and live data prototype
 - d) All of these
34. The values of design thinking are,
 - a) Core values of flexibility, collaboration
 - b) Empathy, Curiosity and positivity
 - c) All of these
 - d) None of these
35. The various types of case studies are,
 - a) Descriptive case study
 - b) Explanatory case study
 - c) Collective case study
 - d) All of these
36. The phases of simulation process are,
 - a) Pre modeling, model building
 - b) Model runs, Experimentation and final thoughts
 - c) All of these
 - d) None of these

37. One of the followings is the type of virtual environment.
 - a) Non-immersive
 - b) Semi-immersive
 - c) Fully-immersive
 - d) All of these
38. The dimensions of collaboration in agile are,
 - a) Collaborative work awareness
 - b) Articulation and appropriation
 - c) All of these
 - d) None of these
39. One of the followings is the method of prototyping,
 - a) Concept sketch, prototype, Bench model prototype
 - b) Fully functional rapid prototype, virtual model prototype
 - c) Pre-production alpha prototype, manufacture production prototype
 - d) All of these
40. The D's of design thinking are,
 - a) Discovery, Design
 - b) Development, Delivery
 - c) All of these
 - d) None of these
41. The following one is the innovative strategies,
 - a) Pro-active and Active
 - b) Reactive and Passive
 - c) All of these
 - d) None of these
42. The strategies used in strategic thinking is,
 - a) Articulate the goals clearly and analyze the possibilities and risks objectively
 - b) Create an implementation plan, adopt and adjust along the way
 - c) All of these
 - d) None of these
43. The storey telling techniques is,
 - a) The conflict
 - b) The character
 - c) The climax
 - d) All of these
44. The elements of strategic management are,
 - a) Arenas, Differentiators, Vehicles
 - b) Stagine and Economic logic
 - c) All of these
 - d) None of these
45. The following are the design strategies,
 - a) Collaboration
 - b) User Centered design, system approach
 - c) Iterative design
 - d) All of these
46. The key components in strategic thinking are,
 - a) Tools for analysis
 - b) Vision and values
 - c) Strategic purposes
 - d) All of these
47. The critical aspects of sense making are,
 - a) Phenomena
 - b) Science and engineering practices
 - c) Students ideas and science ideas
 - d) All of these
48. The types of innovation are,
 - a) Radiant
 - b) Incremental
 - c) Descriptive
 - d) All of these

49. The 3C's of design are,
a) Consistency, Clarity and Content
b) Complete, Complete and Close
c) All of these
d) None of these
50. One of the followings is the teaching principles
a) Integrate new knowledge into the individuals life
b) Test it out in the real world
c) All of these
d) Only (a)

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First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024

ಸಾಂಸ್ಕೃತಿಕ ಕನ್ನಡ

(COMMON TO ALL BRANCHES)

Time: 1 hrs.]

[Max. Marks: 50

ಸೂಚನೆಗಳು

1. ಎಲ್ಲ ಜಂ ಪ್ರಶ್ನೆಗಳಿಗೂ ಉತ್ತರಿಸಿರಿ. ಪ್ರತಿ ಪ್ರಶ್ನೆಗೆ ಒಂದು ಅಂಕ.
2. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಯು.ಎಸ್.ಎನ್ ಸಂಖ್ಯೆ ಹಾಗೂ ಪಶ್ಚಿಮ ಪತ್ರಿಕೆಯ ಶ್ರೇಣಿಯನ್ನು ಅಂದರೆ A, B, C ಅಥವಾ D ಯನ್ನು ತಪ್ಪಿಲ್ಲದಂತೆ ಕಡ್ಡಾಯವಾಗಿ ಗುರುತಿಸುವುದು ಅಭ್ಯರ್ಥಿಯ ಜವಾಬ್ದಾರಿಯಾಗಿರುತ್ತದೆ.
3. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ನಿಗದಿಪಡಿಸಿರುವ ಸ್ಥಳದಲ್ಲಿ ಭರ್ತಿಮಾಡದೆ ಹಾಗೆಯೇ ಬಿಟ್ಟಲ್ಲಿ ಅಥವಾ ಭರ್ತಿಮಾಡಿದ ಮಾಹಿತಿಯಲ್ಲಿ ಯಾವುದೇ ವ್ಯತ್ಯಾಸವಿದ್ದಲ್ಲಿ ಅಂತಹ ಉತ್ತರ ಪತ್ರಿಕೆಗಳನ್ನು ರದ್ದು ಪಡಿಸಲಾಗುವುದು.
4. ಕೇವಲ ಒಂದು ಉತ್ತರವನ್ನು ಮಾತ್ರ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಗುರುತಿಸತಕ್ಕದ್ದು. ಒಂದೆ ಪ್ರಶ್ನೆಗೆ ಎರಡು ಉತ್ತರವನ್ನು ಗುರುತಿಸುವುದು ಅಮಾನ್ಯ.
5. ಎಲ್ಲಾ ಉತ್ತರಗಳನ್ನು ನಿಮಗೆ ಒದಗಿಸಲಾದ ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯ ಹಾಳೆಯ ಮೇಲೆ ಕಪ್ಪು ಅಥವಾ ನೀಲಿ ಶಾಹಿಯ ಬಾಲ್‌ಪಾಯಿಂಟ್ ಪೆನ್ನಿನಿಂದ ಗುರುತು ಮಾಡಬೇಕು.

1. ಕಬ್ಬಿಗರ ಕಾವ್ಯ ಇದರ ಕರ್ತೃ ಯಾರು?
a) ಅಂಡಯ್ಯ b) ಹಂಪನ c) ಪಂಪ d) ಬಸವಣ್ಣ
2. ವಿಜಯನಗರ ಕಾಲದಲ್ಲಿ ಧರ್ಮ ಸಮನ್ವಯತೆಯನ್ನು ಕಾಪಾಡಿದ ರಾಜ ಯಾರು?
a) ಕೃಷ್ಣದೇವರಾಯ b) ಬುಕ್ಕರಾಯ c) ದೇವರಾಯ d) ಹಕ್ಕರಾಯ
3. ಕನ್ನಡ ನಾಡಿನ ಶಾಸನಗಳು ಮುಕ್ತಾಯ ಪಾಲು ಯಾವುದರ ವಿವರಣೆ ಮತ್ತು ಪ್ರಶಸ್ತಿಗೆ ಮೀಸಲಾಗಿದೆ?
a) ದಾನ b) ವೀರ c) ಸಾಹಿತ್ಯ d) ಭಾಷೆ
4. ಅರಸನಿಗೆ ಗಂಡು ಮಗುವಾದರೆ _____ ಕೊಟ್ಟು ಒಬ್ಬ ಪ್ರಾಣ ಬಿಟ್ಟಿದ್ದುಂಟು.
a) ಸಿಡಿತಿಲೆ b) ತಲೆ c) ಹಿಡಿತಿಲೆ d) ಪ್ರಾಣ

5. “ಜಾಗದ ಭೋಗದಕ್ಕರದಗೇಯದ ಗೊಟ್ಟಿಯಲಂಪಿನಿಂಪುಗಳ್ಗೆ ಆಗರವಾದ ಮಾನಸರ್ ” ಎಂಬ ಜೀವನ ರಸಿಕತೆಯ ಆದರ್ಶವನ್ನು ಸಾವಿರ ವರ್ಷಗಳು ಹಿಂದೆ ____ ಹೇಳಿದ್ದಾರೆ.
a) ಆದಿಕವಿ ಪಂಪ b) ರತ್ನಾಕರವರ್ಣಿ c) ಬಸವಣ್ಣ d) ಸರ್ವಜ್ಞ
6. ಸುಮಾರು 1800 ರಲ್ಲಿಯೇ ಕರ್ನಾಟಕ ಏಕೀಕರಣದ ಅಭಿಪ್ರಾಯವನ್ನು ಬ್ರಿಟಿಷ್ ಹಿರಿಯ ಅಧಿಕಾರಿಗಳಿಗೆ ತಿಳಿಸಿದ್ದವರು ಯಾರು.
a) ವಾಲ್ಟರ್ b) ಮಾರ್ಕ್ಸಬ್ಸನ್ c) ಥಾಮಸ್ ಮನ್ರೊ d) ಮೆಕಾಲೆ
7. ಕರ್ನಾಟಕ ಗತವೈಭವ ಗ್ರಂಥದ ಲೇಖಕರು ಯಾರು?
a) ಕುವೆಂಪು b) ಆಲೂರು ವೆಂಕಟರಾಯರು c) ಮಾಸ್ತಿ d) ಜಿ.ಎಸ್. ಎಸ್
8. 1973 ನವೆಂಬರ್ 1 ರಂದು ಮೈಸೂರು ರಾಜ್ಯಕ್ಕೆ ____ ಎಂದು ನಾಮಕರಣ ಮಾಡಲಾಯಿತು.
a) ಮೈಸೂರು b) ನವಕರ್ನಾಟಕ c) ನವಮೈಸೂರು d) ಕರ್ನಾಟಕ
9. ಕನ್ನಡಭಾಷೆ ಯಾವ ಭಾಷಾ ವರ್ಗಕ್ಕೆ ಸೇರಿದೆ?
a) ಸೆಮಿಟಿಕ್ b) ತುರೇನಿಯನ್ c) ದ್ರಾವಿಡ d) ಇಂಡೋಆರ್ಯನ್
10. ಎಷ್ಟನೇ ಅನುಚ್ಛೇದದ ಅನುಸಾರ ಆಯಾ ರಾಜ್ಯಗಳಲ್ಲಿ ಬಳಕೆಯಾಗುವ ಭಾಷೆಗಳನ್ನೇ ಆಡಳಿತ ಭಾಷೆಯೆಂದು ತೀರ್ಮಾನಿಸಲಾಯಿತು?
a) 343 b) 344 c) 345 d) 346
11. ಕನ್ನಡ ಭಾಷೆಯ ಲಿಪಿಯನ್ನು ಲಿಪಿಗಳ ರಾಣಿ ಎಂದು ಕರೆದವರು ಯಾರು?
a) ಸರೋಜಿನಿ ನಾಯ್ಡು b) ಮದರ್ ತೆರೆಸಾ
c) ವಿನೋಬಾ ಭಾವೆ d) ಝಾನ್ಸಿರಾಣಿ ಲಕ್ಷ್ಮೀಬಾಯಿ
12. ಜೇಡರ ದಾಸಿಮಯ್ಯನ ಅಂಕಿತ ಯಾವುದು?
a) ಗುಹೇಶ್ವರ b) ಕೂಡಲ ಸಂಗಮದೇವಾ
c) ರಾಮನಾಥ d) ಚೆನ್ನ ಮಲ್ಲಿಕಾರ್ಜುನ
13. ಹುಲಿಗಂಜಿ ಹುತ್ತವ ಹೊಕ್ಕಡೆ ಏನು ತಿಂಬುದಮಾಬುದೆ?
a) ಸರ್ಪ b) ಆನೆ c) ಕರ್ಮ d) ಮೃತ್ಯು

14. ಕರಿಘನ ಅಂಕುಶ ಕಿರಿಡೆನ್ನಬಹುದೆ ಬಾರದಯ್ಯ ವಚನದ ವಚನಕಾರರು ಯಾರು?
a) ಅಕ್ಕಮಹಾದೇವಿ b) ಆಯ್ದಕ್ಕಿ ಲಕ್ಕಮ್ಮ c) ಆಯ್ದಕ್ಕಿ ಮಾರಯ್ಯ d) ಬಸವಣ್ಣ
15. ಕುರುಡ ಕಣ್ಣ ಕಾಣಲರಿಯದೆ ಏನನ್ನು ಬಯ್ಯನು?
a) ಶಶಿಯ b) ಕನ್ನಡಿಯ c) ಕರ್ಮವ d) ರವಿಯ
16. ಕಾಯಕವೇ ಕೈಲಾಸವಾದ ಕಾರಣ ಅಮರೇಶ್ವರ ಲಿಂಗವಾಯಿತಾದಡು _____ ದೊಳಗು.
a) ಮನಸ್ಸು b) ಕರ್ಮ c) ಕಾಯಕ d) ಕನ್ನಡಿ
17. ತಲ್ಲಣಿಸದಿರು ಕಂಡ್ಯತಾಳು ಮನವೇ ಕೀರ್ತನೆಯ ಕೀರ್ತನಕಾರರು ಯಾರು?
a) ಪುರಂದರದಾಸರು b) ಕನಕದಾಸರು
c) ಶಿಶುನಾಳ ಶರೀಫರು d) ಶಿವಯೋಗಿ
18. ಕುಂಬಾರಕಿ ಮೂರು ಕಾಸಿಗೊಂದು ಏನನ್ನು ಮಾರುತ್ತಾಳೆ?
a) ಕುಡುಕಿ b) ಗಡಿಗೆ c) ಮಡಿಕೆ d) ಗುಡಾಣ
19. ಪುರಂದರದಾಸರ ವಚನಗಳ ಅಂಕಿತನಾಮ ಯಾವುದು?
a) ರಾಮನಾಥ b) ವಿಠಲ c) ಗುಹೇಶ್ವರ d) ಪುರಂದರ ವಿಠಲ
20. ಕುಂಬಾರಕಿ ಮಡಿಕೆ ಬಳಸುವ ಮಣ್ಣನ್ನು ಶರೀಫರು ಯಾವುದಕ್ಕೆ ಹೋಲಿಸಿದ್ದಾರೆ?
a) ಚೆನ್ನ b) ಮುತ್ತು c) ರತ್ನ d) ಲವಣ
21. ಮಂಕುತಿಮ್ಮನ ಕಗ್ಗ ಕಾವ್ಯಭಾಗದ ಕವಿ ಯಾರು?
a) ಬಿ.ಎಂ.ಶ್ರೀ b) ತೀ.ನಂ.ಶ್ರೀ c) ಜಿ.ಎಸ್. ಎಸ್ d) ಡಿವಿಜಿ
22. ಇರುವೆಗಳು ಕಟ್ಟಿದ ಗೂಡು ಯಾವುದಕ್ಕೆ ಹುತ್ತವಾಗುವುದು?
a) ಇಲಿ b) ಬಾವುಲಿ c) ವಿಷನರ್ಪ d) ಮುಂಗುಸಿ
23. ಬೇಂದ್ರೆಯವರ ಯಾವ ಕವನ ಸಂಗ್ರಹಕ್ಕೆ ಜ್ಞಾನಪೀಠ ಪ್ರಶಸ್ತಿ ಬಂದಿದೆ?
a) ಮೂಕಜ್ಜಿಯ ಕನಸುಗಳು b) ನಾಕುತಂತಿ
c) ಸಮಗ್ರ ಸಾಹಿತ್ಯ d) ಸಾಹಿತ್ಯ

24. ಸಾಬಾಣ ಪದದ ಅರ್ಥವೇನು?
 a) ಸಾಬೂನು b) ಸರಕು c) ಸಂಯಮ d) ಯಾವುದು ಅಲ್ಲ
25. ಕಾಲಿಗೆ ಬಿದ್ದವರ ಯಾವುದು ತುಳಿಯುತ್ತಲಿತ್ತು?
 a) ಹಣ b) ಕಾಂಚಾನ c) ದುಡ್ಡು d) ಕುರುಡು ಕಾಂಚಾಣ
26. ಕುವೆಂಪುರವರ ಆತ್ಮಕಥೆ ಯಾವುದು?
 a) ಗಿರಿಜನ b) ನೆನಪಿನ ದೋಣಿ
 c) ಗಿರಿಜನ ಪಯಣ d) ಯಾವುದು ಅಲ್ಲ
27. ಭಗವಂತ ಮಾನವರಿಗೆ ಕಾಣಿಸಿಕೊಂಡರೆ, ಮೊದಲ ಅನ್ನದ ರೂಪದಲ್ಲಿ ಕಾಣಿಸಿಕೊಳ್ಳಬೇಕು ಎಂದು ಹೇಳಿದವರು ಯಾರು?
 a) ಗಾಂಧೀಜಿ b) ನೆಹರು c) ವಿಶ್ವೇಶ್ವರಯ್ಯ d) ಮೂರ್ತಿರಾವ್
28. 56ನೇ ಸನ್ಮದ ಸಾಹಿತ್ಯ ಪರಿಷತ್ತು ಯಾವಾಗ ನಡೆಯಿತು?
 a) 1983 b) 1984 c) 1985 d) 1986
29. ಬಟ್ಟೆಯ ಮೇಲಿನ ಮುದ್ರಣ ಕಲೆಗೆ ಯಾವ ದೇಶವು ಮೂಲ ನೆಲೆಯಾಗಿದೆ?
 a) ಭಾರತ b) ಇಂಗ್ಲೆಂಡ್ c) ಫ್ರಾನ್ಸ್ d) ಅಮೇರಿಕಾ
30. ಡಾ. ಕರೀಗೌಡ ಬೀಚನಹಳ್ಳಿ ಅವರ ಕಾದಂಬರಿ ಯಾವುದು?
 a) ಮಳೆಬಿಲ್ಲು b) ಮಳೆ c) ಕಾಮನಬಿಲ್ಲು d) ಮಳೆಕೋಗಿಲೆ
31. ಭಾರತದಲ್ಲಿ ಬೇರೆ ಬೇರೆ ಬಣ್ಣಗಳನ್ನು ನೀಡುವ ಸುಮಾರು ಎಷ್ಟು ಗಿಡಗಳಿವೆ ಎಂದು ಅಂದಾಜು ಮಾಡಲಾಗಿದೆ?
 a) 200 b) 300 c) 400 d) 500
32. ಎಷ್ಟು ವರ್ಷಗಳಿಂದ ಭಾರತೀಯ ಕರಕುಶಲ ಕಲೆಗಳಿಗೆ ಅಪಾರ ಬೇಡಿದೆ ಇದೆ?
 a) ಕ್ರಿ.ಪೂ.2300 b) ಕ್ರಿ.ಶ.2300 c) ಕ್ರಿ.ಪೂ 2500 d) ಕ್ರಿ.ಶ.2500
33. ಹಸಿವಾದರೆ ಉರೂಳಿಗೆ ಭಿಕ್ಷಾನ್ನಗಳುಂಟು, ತೃಪೆಯಾದರೆ ಕೆರೆ ಬಾವಿ ಹಳ್ಳಗಳುಂಟು, ಶಯನಕ್ಕೆ ಹಾಳು ದೇಗುಲಗಳುಂಟು ಎಂಬ ವಚನ ಬರೆದವರು
 a) ಬಸವಣ್ಣ b) ಅಕ್ಕಮಹಾದೇವಿ c) ಅಲ್ಲಮಪ್ರಭು d) ಜೇಡರ ದಾಸಿಮಯ್ಯ

34. ನಾಕ ಪದದ ಅರ್ಥ
a) ನರಕ b) ಸ್ವರ್ಗ c) ಭೂಮಿ d) ಮಣ್ಣು
35. ವಸುಧೇಂದ್ರ ಅವರ ಕಾದಂಬರಿ ಯಾವುದು?
a) ಹರಿಚಿತ್ತ b) ಸತ್ಯ c) ಚಿತ್ತ d) ಹರಿಚಿತ್ತ ಸತ್ಯ
36. ಗೋಪಣ್ಣ ಮಾಸ್ತರರ ಸ್ನೇಹಿತನ ಹೆಸರೇನು?
a) ಕಾಸಿಂಸಾಬರು b) ಸಾಹೇಬ್
c) ಕರೀಂಖಾನ್ d) ಖಾನ್ ಸಾಹೇಬ್
37. ಪ್ರತಿವರ್ಷ ಮೊಹರಂ ದಿನ ದರ್ಗಾಕ್ಕೆ ಹೋಗಿ ಯಾವ ದೇವರಿಗೆ ಸೇವೆ ಸಲ್ಲಿಸುವ ಸಂಪ್ರದಾಯವನ್ನು ಮಾಸ್ತರರು ರೂಢಿಸಿಕೊಂಡಿದ್ದರು?
a) ಪೀಲು b) ಪೀರ್ c) ಪೀರು d) ಪೀರು
38. ಯುಗಾದಿ ಕಥೆಯಲ್ಲಿ ಬರುವ ದಾದಿಯಾರು?
a) ರಮೆಯ b) ರಾಧಿಕಾ c) ರಾವಣ್ಣ d) ರಾಧಾ
39. ಪ್ರಸ್ತುತ ದಿನಗಳಲ್ಲಿ ಇದನ್ನು ತಯಾರಿಸಲು ಯಂತ್ರಗಳೇ ಇಲ್ಲ.
a) ಬಿದರ ಬುಟ್ಟಿ b) ಬುಟ್ಟಿ c) ಬೆದರಿಬುಟ್ಟಿ d) ಬಿಟ್ಟಿ
40. ಪ್ರಹ್ಲಾದನ ತಾಯಿಯ ಹೆಸರೇನು?
a) ರುದ್ರಮ್ಮ b) ಕಾಳಮ್ಮ c) ರುಕ್ಮಿಣಿ d) ತಾಯಮ್ಮ
41. ಕಾಸಿಂಸಾಬರ ಮಗನ ಹೆಸರೇನು?
a) ಇಸ್ಮೈಲ್ b) ಇಸ್ಮಾಲ್ c) ಇಮ್ರಾನ್ d) ಇಸ್ಮಾಯಿಲ್
42. ರುಕ್ಮಿಣಮ್ಮ ಸತ್ತು ಎಷ್ಟು ವರ್ಷಗಳು ಕಳೆದಿವೆ?
a) 10 b) 15 c) 20 d) 25
43. ಹಾಡುವಳ್ಳಿಗೆ ಹಿಂದೆ ಇದ್ದ ಹೆಸರೇನು?
a) ಸಂಗೀತಪುರ b) ಸಂಗೀತವಳ್ಳಿ
c) ಸಂಗೀತಹಳ್ಳಿ d) ಸಂಗೀತ ನಗರ

44. ಡಾ.ಹಿ.ಚಿ ಬೋರಲಿಂಗಯ್ಯನವರ ಮಾರ್ಗದರ್ಶಕರು ಯಾರು?
a) ಕರಿಯ b) ಕುಪ್ಪಯ್ಯ c) ಕುಪ್ಪಯ್ಯ ಕರಿಯ d) ಕಾಳ
45. ಮೆಗಾನೆ ಎಂಬ ಪ್ರದೇಶದಲ್ಲಿ ವಾಸವಾಗಿರುವ ಜನಾಂಗ ಯಾವುದು?
a) ಕುಣವಿ b) ಕುಣಬಿ c) ಕುಣವ d) ಕುಣಬ
46. ಹಾಡಿನ ಯಜಮಾನ ಯಾರು?
a) ಯಂಕ b) ಯಶು c) ಯಶ d) ಯಂಕು
47. ಯಂಕುವಿನ ಪುರ್ವಜರು ಯಾವ ಕಡೆಯವರು?
a) ಗೋವಾ b) ಕರ್ನಾಟಕ c) ಕೇರಳ d) ತಮಿಳುನಾಡು
48. ಡಾ.ಹಿ.ಚಿ ಬೋರಲಿಂಗಯ್ಯನವರ ಪ್ರವಾಸ ಕಥನಕೃತಿ ಯಾವುದು?
a) ಗಿರಿಜನ b) ಗಿರಿಜನರು
c) ಗಿರಿಜನನಾಡಿಗೆ ಪಯಣ d) ಯಾವುದು ಅಲ್ಲ
49. ಮೆಗಾನೆ ಪರ್ವತದ ಎತ್ತರ
a) 6 ಸಾವಿರ ಅಡಿ b) 5 ಸಾವಿರ ಅಡಿ c) 5 ಕಿಮೀ d) 50 ಅಡಿ
50. ಡಾ.ಹಿ.ಚಿ ಬೋರಲಿಂಗಯ್ಯನವರ ಸ್ನೇಹಿತನ ಹೆಸರೇನು?
a) ನಿಂಗೇಗೌಡ b) ರಾಮೇಗೌಡ c) ಭದ್ರೇಗೌಡ d) ಲಿಂಗೇಗೌಡ

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**First Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Mathematics – I for Civil Engineering Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	With usual notations, prove that $\tan \phi = r \frac{d\theta}{dr}$.	6	L2	CO1
	b.	Find the angle between the curves $r = a(1 + \sin\theta)$ and $r = a(1 - \sin\theta)$.	7	L2	CO1
	c.	Show that the radius of curvature for the curve $r^n = a^n \cos n\theta$ varies inversely as r^{n-1} .	7	L3	CO1
OR					
Q.2	a.	Show that the radius of curvature for the catenary of uniform strength $y = a \log[\sec(x/a)]$ is a $\sec(x/a)$.	7	L2	CO1
	b.	Find the pedal equation of the curve : $\frac{2a}{r} = (1 + \cos\theta)$.	8	L2	CO1
	c.	Using modern mathematical tool, write a program/code to plot the sine and cosine curve.	5	L3	CO5
Module – 2					
Q.3	a.	Expand $\log(\sec x)$ upto the terms containing x^6 using Maclaurin's series.	6	L2	CO1
	b.	If $u = f(x - y, y - z, z - x)$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.	7	L3	CO1
	c.	Examine the function $f(x, y) = xy(a - x - y)$ for extreme values.	7	L3	CO1
OR					
Q.4	a.	If $u = e^{ax+by} \times f(ax - by)$, prove that $b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y} = 2abu$.	7	L2	CO1
	b.	If $u = x + 3y^2 - z^3$, $v = 4x^2yz$, $w = 2z^2 - xy$, find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ at $(1, -1, 0)$.	8	L3	CO1
	c.	Using modern mathematical tool write a program/code to evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$.	5	L3	CO5

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Module – 3

Q.5	a.	Solve: $\frac{dy}{dx} + \frac{y}{x} = y^2x$	6	L2	CO2
	b.	Show that the family of parabolas $y^2 = 4a(x + a)$ is self orthogonal.	7	L3	CO2
	c.	Solve : $p^2 + 2py \cot x = y^2$.	7	L2	CO2

OR

Q.6	a.	Solve : $(2x + y + 1) dx + (x + 2y + 1) dy = 0$.	6	L2	CO2
	b.	A body originally at 80°C cools down to 60°C in 20 minutes, the temperature of the air being 40°C. What will be the temperature of the body after 40 minutes from the original?	7	L3	CO2
	c.	Solve the equation $(px - y)(py + x) = 2p$ by reducing into Clairaut's form, taking the substitution $X = x^2$, $Y = y^2$.	7	L2	CO2

Module – 4

Q.7	a.	Solve : $(4D^4 - 4D^3 - 23D^2 + 12D + 36) y = 0$.	6	L2	CO3
	b.	Solve : $(D - 2)^2 y = 8(e^{2x} + \sin 2x)$.	7	L2	CO3
	c.	Solve by the method of variation of parameters $y'' + a^2y = \sec ax$.	7	L3	CO3

OR

Q.8	a.	Solve : $y'' + 3y' + 2y = 12x^2$.	6	L2	CO3
	b.	Solve : $\frac{d^2y}{dx^2} - 4y = \cosh(2x - 1) + 3^x$.	7	L2	CO3
	c.	Solve : $(2x + 1)^2 y'' - 6(2x + 1)y' + 16y = 8(2x + 1)^2$.	7	L3	CO3

Module – 5

Q.9	a.	Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 2 & 4 & 6 & 8 \\ 4 & 8 & 12 & 16 \\ 1 & 2 & 3 & 4 \end{bmatrix}$.	6	L2	CO4
	b.	Solve the system of equations by using Gauss-Jordan method: $2x + y + z = 10$, $3x + 2y + 3z = 18$, $x + 4y + 9z = 16$.	7	L3	CO4
	c.	Using Rayleigh's power method, find the dominant eigen value and corresponding eigen vector of $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by taking $[1, 1, 1]^T$ as the initial eigen vector. Carryout 6 iterations.	7	L3	CO4

OR

Q.10	a.	Find the rank of the matrix $\begin{bmatrix} 11 & 12 & 13 & 14 \\ 12 & 13 & 14 & 15 \\ 13 & 14 & 15 & 16 \\ 14 & 15 & 16 & 17 \end{bmatrix}$	7	L3	CO4
	b.	Solve the system of equations by using Gauss-Seidel method: $10x + 2y + z = 9$, $x + 10y - z = -22$, $-2x + 3y + 10z = 22$. Carryout 3 iterations.	8	L3	CO4
	c.	Using modern mathematical tool, write a program/code to test the consistency of the equations $x + 2y - z = 1$, $2x + y + 4z = 2$, $3x + 3y + 4z = 1$.	5	L3	CO5

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**First Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Mathematics – I for EEE Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. *M* : Marks , *L*: Bloom's level , *C*: Course outcomes.

Module – 1			M	L	C
Q.1	a.	With usual notations prove that $\cot \phi = \frac{1}{r} \left(\frac{dr}{d\theta} \right)$	6	L2	CO1
	b.	Find the angle between the curves $r = 6 \cos \theta$ and $r = 2(1 + \cos \theta)$	7	L2	CO1
	c.	Find the radius of curvature of the curve $\sqrt{x} + \sqrt{y} = 4$ at the point where it cuts the line passing through the origin making an angle 45° with the X-axis.	7	L3	CO1
OR					
Q.2	a.	Find the Pedal equation of the curve $r^2 = a^2 \sec 2\theta$	7	L2	CO1
	b.	Show that for the curve $r = a(1 + \cos \theta)$ is $p^2/r = \text{constant}$.	8	L3	CO1
	c.	Using modern mathematical tool, write a program/code to plot the sine and cosine curve.	5	L3	CO5
Module – 2					
Q.3	a.	Expand $\log(\sec x)$ up to the term containing x^4 using Maclaurin's series.	6	L2	CO1
	b.	If $u = f(x - y, y - z, z - x)$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$	7	L2	CO1
	c.	If $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$ find the value of $\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)}$	7	L3	CO1
OR					
Q.4	a.	If $z = e^{ax+by} f(ax-by)$, show that $b \frac{\partial z}{\partial x} + a \frac{\partial z}{\partial y} = 2abz$	7	L2	CO1
	b.	Find the extreme values of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$	8	L3	CO1

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	c.	Using modern mathematical tool, write a program/code to evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$	5	L3 CO5
Module – 3				
Q.5	a.	Solve $\frac{dy}{dx} + \frac{y}{x} = y^2x$	6	L2 CO2
	b.	Find the orthogonal trajectories of the family of curves $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \alpha} = 1$, where α is a parameter.	7	L3 CO2
	c.	Solve $xyp^2 - (x^2 + y^2)p + xy = 0$.	7	L2 CO3
OR				
Q.6	a.	Solve $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$	6	L2 CO2
	b.	Show that a differential equation for the current i in an electrical circuit containing an inductance L and resistance R in series and acted on by an electromotive force $E \sin \omega t$, satisfies the equation $L \frac{di}{dt} + Ri = E \sin \omega t$. Find the value of the current at any time t , if initially there is no current in the circuit.	7	L3 CO2
	c.	Solve the equation $(px - y)(py + x) = 2p$ by reducing in to Clairaut's form, taking the substitution $X = x^2$, $Y = y^2$.	7	L2 CO2
Module – 4				
Q.7	a.	Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x + y + z) dy dx dz$	6	L2 CO3
	b.	Evaluate $\int_{-2}^2 \int_0^{\sqrt{4-x^2}} (2-x) dy dx$ by changing the order of integration.	7	L2 CO3
	c.	Prove that $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \times \int_0^{\pi/2} \sqrt{\sin \theta} d\theta = \pi$	7	L2 CO3
OR				
Q.8	a.	Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing into polar coordinates.	6	L2 CO3
	b.	Derive the relation between beta and gamma function.	7	L2 CO3
	c.	Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$, $z = 0$.	7	L3 CO3
2 of 3				

Module – 5

Q.9	a.	Find the rank of the matrix $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$	6	L2	CO4
	b.	Solve the system of equations by Gauss elimination method $2x + y + 4z = 12$ $4x + 11y - z = 33$ $8x - 3y + 2z = 20$	7	L3	CO4
	c.	Solve the following system of equations by Gauss-Seidel method $20x + y - 2z = 17$ $3x + 20y - z = -18$ $2x - 3y + 20z = 25$	7	L3	CO4
OR					
Q.10	a.	Using Gauss Jordan method, solve $2x + 5y + 7z = 52, 2x + y - z = 0, x + y + z = 9$	7	L3	CO4
	b.	Using Rayleigh's power method find the dominant eigenvalue and the corresponding eigenvector of $\begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ by taking $[1, 0, 0]^T$ as initial eigen vector [carryout 5 iterations].	8	L3	CO5
	c.	Using modern mathematical tool. Write a program/code to test the consistency of equations: $x + 2y - z = 1$ $2x + y + 4z = 2$ $3x + 3y + 4z = 1$	5	L3	CO5

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**First Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Mathematics – I for ME Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. M : Marks, L : Bloom's level, C : Course outcomes.

Module – 1			M	L	C
Q.1	a.	With usual notations prove that $\tan \phi = r \cdot \frac{d\theta}{dr}$.	6	L2	CO1
	b.	Find the angle of intersection for the pair of curves $r = a(1 + \cos \theta)$, $r = b(1 - \cos \theta)$.	7	L2	CO1
	c.	Find the Pedal equation for the curve $r^n = a^n \cos n\theta$.	7	L2	CO1
OR					
Q.2	a.	Derive an expression for radius of curvature in Cartesian form $\int = \frac{[1 + y_1^2]^{3/2}}{y_2}$.	7	L1	CO1
	b.	Show that the pair of curves intersect each other orthogonally. $r^n = a^n \cos n\theta$ and $r^n = b^n \sin n\theta$.	8	L3	CO1
	c.	Using modern mathematical tool write a program to plot sine and cosine curve.	5	L3	CO5
Module – 2					
Q.3	a.	Expand $\log(1 + x)$ upto the term containing x^4 , using Maclaurins series.	6	L2	CO2
	b.	If $u = f(p, q, r)$ where $p = x-y$, $q = y-z$, $r = z-x$. Show that, $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$	7	L2	CO2
	c.	Show that $z(x, y) = x^3 + y^3 - 3xy + 1$ is minimum at $(1, 1)$.	7	L3	CO2
OR					
Q.4	a.	Evaluate the $\lim_{x \rightarrow 0} \left[\frac{a^x + b^x + c^x}{3} \right]^{1/x}$.	8	L2	CO2
	b.	If $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$ find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$.	7	L2	CO2

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	c.	Using modern mathematical tool, write a program/code to evaluate: $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$.	5	L5	CO5
Module – 3					
Q.5	a.	Solve the Bernoulli's differential equation: $\frac{dy}{dx} + \frac{y}{x} = y^2 x$.	6	L2	CO3
	b.	Find the orthogonal trajectories of the family $y^2 = cx^3$.	7	L3	CO3
	c.	Solve : $y \cdot p^2 + (x - y) p - x = 0$.	7	L2	CO3
OR					
Q.6	a.	Solve : $(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0$.	6	L2	CO3
	b.	A body in air at 25°C cools from 100°C to 75°C in 1 minute. Find the temperature of the body at the end of 3 minutes.	7	L3	CO3
	c.	Modify the equation into Clairaut's form. Hence find the general and singular solution of $xp^2 - py + kp + a = 0$.	7	L2	CO3
Module – 4					
Q.7	a.	Solve : $(D^3 - 2D^2 + 4D - 8)y = 0$.	6	L2	CO3
	b.	Solve : $(6D^2 + 17D + 12)y = e^{-x}$.	7	L2	CO3
	c.	Solve by variation of parameters $(D^2 + 1)y = \tan x$.	7	L2	CO3
OR					
Q.8	a.	Solve : $(D^3 + 1)y = 0$	6	L2	CO3
	b.	Solve : $y'' + 2y' + y = 2x + x^2$	7	L2	CO3
	c.	Solve : $x^2y'' - 2y = x^2$	7	L2	CO3
Module – 5					
Q.9	a.	Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 2 & 3 & 5 & 1 \\ 1 & 3 & 4 & 5 \end{bmatrix}$	6	L2	CO4
	b.	Solve the system of equation by using Gauss elimination method $x + 2y + z = 3$, $2x + 3y + 3z = 10$, $3x - y + 2z = 13$	7	L3	CO4

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	c.	Using Rayleigh's power method, find the largest eigen value and the corresponding eigen vector of the matrix, $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by taking initial vector as $[1 \ 1 \ 1]^T$.	7	L3 CO4
OR				
Q.10	a.	Solve the system of equation by using Gauss-Jordan method. $x + y + z = 8, -x - y + 2z = -4, 3x + 5y - 7z = -14$	7	L3 CO4
	b.	Solve the system of equations by Gauss-Seidel method $20x + y - 2z = 17, 3x + 20y - z = -18, 2x - 3y + 20z = 25.$	8	L3 CO4
	c.	Using modern mathematical tool write a program/code to test the consistency of the equations $x + 2y - z = 1, 2x + y + 4z = 2, 3x + 3y + 4z = 1.$	5	L3 CO5

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BMATS101

**First Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Mathematics – I for CSE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	With usual notations, prove that $\tan \phi = r \frac{d\theta}{dr}$.		6	L2	CO1
	b.	Show that the curves $r = a(1 + \sin\theta)$ and $r = b(1 - \sin\theta)$ intersect each other orthogonally.		7	L2	CO1
	c.	Find the radius of curvature at the point $\left(\frac{39}{2}, \frac{39}{2}\right)$ for the curve $x^3 + y^3 = 3axy$.		7	L3	CO1
OR						
Q.2	a.	With usual notations prove that $\rho = \frac{[1 + y_1^2]^{\frac{3}{2}}}{y_2}$		8	L2	CO1
	b.	Find the pedal equation of the curve : $r^n = a^n \cos n\theta$.		7	L1	CO1
	c.	Using modern mathematical tool, write a program/code to plot the curve $r = 2 \cos 2\theta $.		5	L1	CO5
Module – 2						
Q.3	a.	Obtain the Maclaurin's expansion of $\log(1 + e^x)$ upto the term containing x^4 .		6	L2	CO1
	b.	If $u = f[2x - 3y, 3y - 4z, 4z - 2x]$ then find the value of $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z}$.		7	L1	CO1
	c.	Find the maximum and minimum value of the function, $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$.		7	L1	CO1
OR						
Q.4	a.	Evaluate $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x + d^x}{4} \right)^{\frac{1}{x}}$		7	L2	CO1

	b.	If $u = x^2 - y^2$, when $x = e^t \cos t$, $y = e^t \sin t$, show that $\frac{\partial u}{\partial t} = 2e^{2t}[\cos 2t - \sin 2t]$.	8	L2	CO1
	c.	Using modern mathematical tool write a program/code to show that $u_{xx} + u_{yy} = 0$, given $u = e^x[x \cos y - y \sin y]$.	5	L3	CO5
Module – 3					
Q.5	a.	Solve: $\frac{dy}{dx} + \frac{y}{x} = y^2 x$	6	L3	CO2
	b.	Find the orthogonal trajectories of $\frac{x^2}{a^2 + x} + \frac{y^2}{b^2 + x} = 1$ where λ is a parameter.	7	L1	CO2
	c.	Solve : $x^2 p^2 + xyp - 6y^2 = 0$.	7	L3	CO2
OR					
Q.6	a.	Solve : $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$.	6	L2	CO2
	b.	Solve the differential equation $L \frac{di}{dt} + Ri = 200 \sin 300t$ when $L = 0.05$ and $R = 100$ and find the value of the current I at any time t , it initially there is no current in the circuit. What value does i approach after a long time.	7	L3	CO2
	c.	Find the general and singular solution of $(a^2 - x^2)p^2 + 2xyp + b^2 - y^2 = 0$	7	L2	CO2
Module – 4					
Q.7	a.	(i) Find the remainder when 41^{75} is divided by 3, (ii) Find the last digit in 7^{289} .	6	L2	CO3
	b.	Find the solution of the linear congruence $18x \equiv 30 \pmod{42}$.	7	L2	CO3
	c.	Using RSA algorithm find public key and private key with respect to $p = 3$, $q = 11$ and $m = 31$	7	L3	CO3
OR					
Q.8	a.	Show that $8^{30} - 1$ is divisible by 31 using Fermat to little theorem.	6	L2	CO3
	b.	Solve the system of linear congruence's using CRT $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$.	7	L3	CO3
	c.	(i) Find the remainder when $349 \times 74 \times 36$ is divided by 3. (ii) Find the roots of $x^2 + 2x - 3 \equiv 0 \pmod{5}$	7	L2	CO3
Module – 5					
Q.9	a.	Find the rank of the matrix $\begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$.	6	L2	CO4

	b.	Test for consistency and solve $5x + 3y + 7z = 4$, $3x + 26y + 2z = 9$, $7x + 2y + 10z = 5$	7	L3	CO4
	c.	Find the largest Eigen value and the corresponding Eigen vector, $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ with initial vector $[1 \ 1 \ 1 \ 1]^T$ carry out 5 iterations.	7	L1	CO4
OR					
Q.10	a.	Solve the system of equations by Gauss Seidal method $83x + 11y - 4z = 95$, $7x + 52y + 13z = 104$, $3x + 8y + 29z = 71$. Carry out three iterations.	8	L4	CO3
	b.	Solve the system of equations by using Gauss-Jordan method: $x + 2y + z = 8$ $2x + 3y + 4z = 20$, $4x + 3y + 2z = 16$	7	L3	CO4
	c.	Using modern mathematical tool, write a program/code to find the largest eigen value of, $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$	5	L3	CO5

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BPHYE102/202

First/Second Semester B.E./B.Tech. Degree Supplementary Examination, June/July 2024 Applied Physics for EEE Stream

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Obtain the expression for Eigen value and Eigen function for a particle in one dimensional potential well of infinite height.	10	L2	CO1
	b.	What is Wave function? Give its physical significance and properties.	6	L2	CO1
	c.	An electron is bound in an one dimensional potential well of width $1\text{\AA}$ and of infinite height. Find its energy values in eV in the ground state and first two excited states.	4	L3	CO1
OR					
Q.2	a.	State and explain Heisenberg's uncertainty principle. Show that an electron does not exist inside the nucleus on the basis of Heisenberg's uncertainty principle.	10	L2	CO1
	b.	What is wave function, probability density and normalization of wave function.	6	L2	CO1
	c.	Calculate the de Broglie wavelength associated with an electron having a kinetic energy of 100eV.	4	L3	CO1
Module – 2					
Q.3	a.	Describe Type I and Type II superconductors. Explain construction and working of MAGLEV vehicle.	10	L2	CO2
	b.	Derive Clausius Mossotti equation.	6	L2	CO2
	c.	Show that the sum of the probability of occupancy of an energy state of ΔE below Fermi level and that at ΔE above Fermi level is unity.	4	L3	CO2
OR					
Q.4	a.	Discuss the probability of occupation of various energy states by electrons at $T = 0\text{K}$ and $T > 0\text{K}$ on the basis of fermi factor.	10	L2	CO2
	b.	Describe different types of polarization mechanisms in dielectric materials.	6	L2	CO2
1 of 3					

	c.	If NaCl crystal is subjected to an electric field of 1000V/m and the resulting polarization is $4.3 \times 10^{-8} \text{ C/m}^2$. Calculate dielectric constant of NaCl.	4	L3	CO2
Module – 3					
Q.5	a.	What is numerical aperture? Obtain an expression for numerical aperture of optical fiber.	8	L2	CO3
	b.	Describe the construction and working of carbon dioxide laser with energy level diagram.	8	L2	CO3
	c.	In a diffraction grating experiment with a semiconductor laser, the grating constant is 4.7×10^{-5} per metre. If the angle of diffraction for the second order is 1.5 degree, find the wavelength of the laser.	4	L3	CO5
OR					
Q.6	a.	Obtain an expression for energy density of radiation under thermal equilibrium condition in terms of Einstein's coefficients.	10	L2	CO3
	b.	Explain different types of attenuations of fibers.	6	L2	CO3
	c.	Determine the resonance frequency of an LCR series circuit with inductance = 0.5H, capacitance = 0.45 microfarad and resistance $300 = \Omega$.	4	L3	CO5
Module – 4					
Q.7	a.	Derive wave equation for electromagnetic waves in vacuum in terms of electric field using Maxwell's equation.	9	L2	CO4
	b.	Explain the terms gradient of scalar, divergence and curl of a vector.	6	L2	CO4
	c.	Prove that $3y^4z^2\hat{a}_x + 4x^3z^2\hat{a}_y + 3x^2y^2\hat{a}_z$ is solenoidal.	5	L3	CO4
OR					
Q.8	a.	State and prove Gauss Divergence theorem.	7	L2	CO4
	b.	Explain Faraday's law of electromagnetic induction. Express the same in the point form of Maxwell's equation.	8	L2	CO4
	c.	Find the divergence of the vector field $\vec{A}$ given by $\vec{A} = 6x^2\hat{a}_x + 3xy^2\hat{a}_y + xyz^3\hat{a}_z$.	5	L3	CO4
Module – 5					
Q.9	a.	What is Hall effect? Obtain an expression for the Hall coefficient in terms of Hall voltage.	8	L2	CO5
	b.	Explain the construction and working of photodiode. Define the power responsivity in a photodiode.	8	L2	CO5
	c.	The resistivity of intrinsic Germanium at 27°C is equal to 0.47 ohm-meter. Assuming electron and hole mobilities are $0.38 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $0.18 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ respectively. Calculate the intrinsic carrier density.	4	L3	CO5

OR

Q.10	a.	Establish relation between Fermi energy and energy gap for an intrinsic semi conductor.	8	L2	CO5
	b.	Derive expression for electrical conductivity in extrinsic and intrinsic semiconductors.	8	L2	CO5
	c.	The following data are given for intrinsic Germanium at 300K, $n_i = 2.4 \times 10^{19}/\text{m}^3$, $\mu_e = 0.39\text{m}^2\text{v}^{-1}\text{s}^{-1}$, $\mu_h = 0.19\text{m}^2\text{v}^{-1}\text{s}^{-1}$. Calculate the resistivity of the sample.	4	L3	CO5

First/Second Semester B.E./B.Tech Degree Supplementary Examination, June/July 2024

Applied Physics for ME Stream

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define spring constant. Obtain expression for equivalent force constant for two springs connected in series and parallel combination.	9	L2	CO1
	b.	Obtain a differential equation for a body undergoing forced oscillation and mention expression for amplitude and phase of oscillation.	6	L2	CO1
	c.	A mass of 0.5kg causes an extension of 0.03m in a spring and the system is set for oscillations. Find the force constant of the spring, angular frequency and the time period of the resulting oscillations.	5	L3	CO2
OR					
Q.2	a.	Describe the construction and working of hand operated Reddy shock tube. Mention any two key feature of Reddy shock tube.	10	L2	CO1
	b.	Discuss the conditions for resonance and explain the sharpness of resonance.	6	L2	CO1
	c.	The distance between the two pressure sensors in a shock tube is 150mm. The time taken by a shock wave to travel the distanced is 0.3ms, if the velocity of sound under the same condition is 340ms^{-1} . Find the Mach number of the shock wave.	4	L3	CO1
Module – 2					
Q.3	a.	Define Young's modulus, bulk modulus and rigidity modulus. Derive relation between y , n and σ .	10	L2	CO1
	b.	With neat diagram explain the stress-strain curve for elastic materials.	6	L2	CO1
	c.	Calculate the Poisson's ratio for the material given that $y = 12.25 \times 10^{10}\text{N/m}^2$ and $\eta = 4.55 \times 10^{10}\text{N/m}^2$.	4	L3	CO2
OR					
Q.4	a.	Explain the term bending moment. Show the bending moment of a thin uniform bar of rectangular cross section is $\frac{Ybd^3}{12R}$.	10	L2	CO1
	b.	What is the fracture of elastic materials? Discuss on ductile and brittle fractures.	6	L2	CO1
	c.	Calculate the force required to procedure an extension of 1mm in steel wire of length 2m and diameter 1mm [Young's modulus for steel is $2 \times 10^{11}\text{N/m}^2$].	4	L3	CO1
1 of 3					

Module – 3

Q.5	a.	Discuss seebeck effect and peltier effect with their co-efficient.	8	L2	CO2
	b.	State and explain laws of thermo electricity's.	8	L2	CO2
	c.	The emf in lead-iron thermocouple, one junction of which is at 0°C, is given by $E = 1784t - 2.4t^2$ (in μ volts) where t is temperature in °C. Find the neutral temperature.	4	L3	CO2

OR

Q.6	a.	Derive expression for thermo emf in terms of T_1 and T_2 .	8	L2	CO2
	b.	What are thermoelectric materials? Explain low, mid and high temperature thermoelectric materials.	8	L2	CO2
	c.	For Fe – Cu thermocouple it is observed that the thermo emf is zero when one of the junctions is at 20°C and other one is at some higher temperature. If the neutral temperature is 285°C, Calculate the higher temperature. Hence find out the temperature of inversion, if the cold junction temperature is at – 20°C.	4	L3	CO2

Module – 4

Q.7	a.	Explain Joule Thomson effect show that, $\Delta T = \frac{(P_1 - P_2)}{C_p} \left[\frac{2a}{RT} - b \right]$	10	L2	CO3
	b.	Explain the liquefaction of Helium.	6	L2	CO3
	c.	Calculate inversion temperature of gas. Given $a = 0.244 \text{ atm L}^2/\text{mol}^2$, $b = 0.027 \text{ L/mol}$ and $R = 0.0821 \text{ L atm/K/mol}$.	4	L3	CO3

OR

Q.8	a.	Explain the construction and working of porous plug experiment with neat diagram.	10	L2	CO3
	b.	Explain briefly the applications of cryogenics in food processing and a aerospace.	6	L2	CO3
	c.	In Joule Thomson experiment temperature changes from 100°C to 150°C for pressure change of 20MPa to 170MPa. Calculate Joule Thomson co-efficient.	4	L3	CO3

Module – 5

Q.9	a.	With neat diagram, explain the construction and working of x-ray diffractometer.	10	L2	CO4
	b.	Define nano-material and nano-composite and classify the nano-materials based on the dimensional constraints.	6	L2	CO4
	c.	First order Bragg reflection occurs when a monochromatic beam of x-rays of wavelength $0.675 \text{ \AA}$ is incident on a crystal at a glancing angle of 4.85° . What is the glancing angle for third order Bragg reflection to occur?	4	L3	CO4
OR					
Q.10	a.	Describe the construction and working of atomic force microscopy.	8	L2	CO4
	b.	Give the principle, construction and working of Scanning Electron Microscope (SEM).	8	L2	CO4
	c.	Determine the crystallite size given the wavelength of x-rays 10 nm , the peak width 0.5° and peak position 25° for a cubic crystal given $K = 0.94$.	4	L3	CO4

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CBCS SCHEME

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BPHYS102/202

**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Applied Physics for CSE Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Obtain the expression for energy density using Einstein's coefficients of absorption and emission.	08	L2	CO1
	b.	Define angle of acceptance. Explain types of optical fibres with neat ray diagram and refractive index profile.	07	L2	CO1
	c.	Find the ratio of population of the two states in a He-Ne laser that produce light of wavelength 6328 Å at 27°C.	05	L3	CO1
OR					
Q.2	a.	Derive an expression for numerical aperture of an optical fibre in terms of fraction of RI.	08	L2	CO1
	b.	What is active medium? Explain construction and working of semiconductor laser.	07	L2	CO1
	c.	For a 30 cm long fiber attenuation 0.8 dB/km, find the output power if a 200 μW power is launched?	05	L3	CO1
Module – 2					
Q.3	a.	Define group velocity, phase velocity, wave function and probability density.	08	L2	CO2
	b.	Derive time-independent Schrödinger wave equation.	07	L2	CO2
	c.	Which has shorter wavelength, a 10 eV photon or a 10 eV electron? Explain.	05	L2	CO2
OR					
Q.4	a.	State and explain Heisenberg's uncertainty principle. Explain principle of complementarity.	07	L2	CO2
	b.	Derive the wave function of a particle inside infinite potential well of width 'a' using normalization condition.	08	L2	CO2
	c.	Compute the energy of the lowest three levels for an electron in a square well of width 3 Å.	05	L3	CO2
Module – 3					
Q.5	a.	What is Bloch Sphere? Represent $ 0\rangle$ and $ 1\rangle$ on the Bloch sphere.	08	L2	CO1
	b.	Explain probability, normalization and quantum superposition.	07	L2	CO1
	c.	Using two X-gates in series, show that two not gates in series are equivalent to a quantum wire.	05	L2	CO1
OR					
Q.6	a.	Discuss two qubit quantum NOT gate or controlled NOT gate with four different input states.	08	L2	CO3
	b.	State Moore's law. Show that S gate can be formed by connecting two T gates in series.	07	L3	CO3
	c.	Find the inner product basis of states $ 1\rangle$ and $ 0\rangle$, and draw conclusions on the result.	05	L3	CO3

Module – 4

Q.7	a.	State Mathiessen's rule. Discuss variation of Fermi factor with temperature and energy.	08	L2	CO4
	b.	Define critical field. Write a note on high temperature super conductors.	07	L1	CO4
	c.	In a solid, consider the energy level lying 0.01 eV below the fermilevel. What is the probability of this level not being occupied by an electron?	05	L3	CO4

OR

Q.8	a.	Define super conductors. Give brief account on BCS theory of super conductors.	08	L2	CO4
	b.	What is density of states? Explain failures of classical free electron theory.	07	L2	CO4
	c.	Find the transition temperature of a metal whose critical magnetic field is 5×10^3 A/m at 6 K and 2×10^4 A/m at 0 K.	05	L3	CO4

Module – 5

Q.9	a.	What are frames and frames per seconds? Explain how the odd rule can be applied to place the object in specific frames.	08	L2	CO5
	b.	Explain Monte-Carlo method applied to approximating the value of x.	07	L2	CO5
	c.	While animating a speeding up car the total distance travelled over 6 frames is 25 m, calculate the basic distance.	05	L2	CO5

OR

Q.10	a.	Define jump magnification. Explain how to calculate jump timing.	07	L2	CO5
	b.	Explain Poisson and normal distribution with their probability functions.	08	L2	CO5
	c.	The number of particles emitted per second by a random radioactive source has a Poisson distribution with $\alpha = 4$. Calculate the probability of $P(x = 0)$, $P(x = 1)$ and $P(x = 2)$.	05	L3	CO5

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OR

Q.6	a.	Explain any 5 methods in os.path module related to files.	10	L2	CO3
	b.	Explain file reading and writing process with example.	10	L3	CO3

Module – 4

Q.7	a.	Write a program to display folder name, list of subfolders, and files in the working directory using os.walk().	5	L3	CO3
	b.	Explain the following with respect to shutil module. i) Copying files and folders ii) Moving and renaming files and folders.	8	L3	CO3
	c.	Write a program to backup a folder into a ZIP file.	7	L3	CO3

OR

Q.8	a.	What is an assertion? Explain how to use assert keyword with an example.	7	L3	CO3
	b.	Explain the different logging levels.	7	L2	CO3
	c.	Demonstrate reading and extracting from zip files using zipfile module.	6	L3	CO3

Module – 5

Q.9	a.	Explain __init__, __str__, __add__ methods with example.	12	L3	CO4
	b.	Explain type based dispatch with example.	8	L3	CO4

OR

Q.10	a.	Define classes and objects. Write a program to create a class called student with attributes name, usn, sem, sec and create two student objects. Read and print the details of two students using appropriate methods.	12	L3	CO4
	b.	Explain pure functions with examples.	8	L3	CO4

CBCS SCHEME

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BPLCK205B/BPLCKB205

**Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Introduction to Python Programming

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. *M* : Marks , *L*: Bloom's level , *C*: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Explain elif, for, while statement in python with example for each.	06	L2	CO1
	b.	List and explain math operators used in python with example.	06	L2	CO1
	c.	Develop a program to read the name and year of birth of a person. Print whether person is senior citizen or not.	08	L3	CO1
OR					
Q.2	a.	Explain local and global scope with example.	06	L2	CO1
	b.	With an example, explain the following built in function : (i) print() (ii) input() (iii) len()	06	L2	CO1
	c.	Develop a program to generate Fibonacci number of length (N). Read N from the console.	08	L3	CO1
Module – 2					
Q.3	a.	Explain the following list methods with example : (i) append() (ii) insert() (iii) sort()	08	L2	CO2
	b.	Differentiate List and dictionaries.	04	L1	CO2
	c.	Develop a program using dictionary to print Ten most frequently appearing word in a text file.	08	L3	CO2
OR					
Q.4	a.	Explain the following method with example: (i) key() (ii) values() (iii) items() in dictionary	08	L2	CO2
	b.	Show that List are Mutable.	04	L1	CO2
	c.	Develop a program to compute Mean, Variance, Standard deviation with message.	08	L3	CO2
Module – 3					
Q.5	a.	Explain the following string method with example: (i) isalpha() (ii) isalnum() (iii) isdecimal() (iv) isspace()	08	L1	CO3

1 of 2

	b.	Differentiate between absolute and relative path in specify file path.	04	L2	CO3
	c.	Write a program to accept string and display total number of alphabet.	08	L3	CO3
OR					
Q.6	a.	Explain the following method with example: (i) upper() (ii) lower() (iii) is_upper() (iv) is_lower()	08	L2	CO3
	b.	Explain how to save variable with Shelve module.	04	L2	CO3
	c.	Develop a program to sort the content of a text file and write the sorted content into separate file.	08	L2	CO3
Module – 4					
Q.7	a.	How do you copy files and folders using Shutil module? Explain in detail.	10	L2	CO3
	b.	With suitable code, explain Backup a folder into a Zip files, clearly mention steps in detail.	10	L3	CO3
OR					
Q.8	a.	What are assertions? Write the content of an assert statement. Explain then with example.	10	L2	CO3
	b.	Explain logging module with example how files and folder can be permanently deleted.	10	L2	CO3
Module – 5					
Q.9	a.	What is a class? How to define class in python? How to initiate a class and how the class members are accessed?	10	L2	CO4
	b.	What is polymorphism? Demonstrate polymorphism with function to find histogram to count the number of times each letter appears in a word and in sentences.	10	L3	CO4
OR					
Q.10	a.	Discuss operator overloading. Mention any five operators with respective special function to be overloaded in python.	10	L2	CO4
	b.	Define pure function. Illustrate with an example.	10	L3	CO4

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BPLCK205C/ BPLCKC 205

**Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Basics of Java Programming

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Describe the structure of Java Programming.	6	L1	CO1
	b.	Explain in detail, about the four groups of primitive types of data.	10	L1	CO1
	c.	Write Java program to initialize and display different type of integer and floating point variables.	4	L2	CO1
OR					
Q.2	a.	Explain Type Conversion with an example.	6	L2	CO1
	b.	Explain different ways of array declaration with syntax.	6	L2	CO1
	c.	Explain the concept of object oriented programming.	8	L2	CO1
		(i) Encapsulation			
(ii) Inheritance					
(iii) Polymorphism					
	(iv) Data Abstraction				
Module – 2					
Q.3	a.	Explain the operation of the following operator with example.	8	L2	CO3
		(i) %			
		(ii) >>>			
		(iii) &&			
		(iv) ?			
b.	Write a JAVA program to sort list of elements in ascending and descending order.	12	L3	CO1	
OR					
Q.4	a.	Write a program to calculate the average among the element {8, 4, 1, 2} using ‘for each’ in JAVA. How for each is differ from for?	10	L3	CO1
	b.	List and explain Java Selection Statements in detail with example.	10	L2	CO1
Module – 3					
Q.5	a.	What is constructor and parameterized constructors? Explain in detail with example. Write a program for constructor to use ‘this’ keyword.	10	L2	CO1
	b.	Develop a JAVA program demonstrating method overloading and constructor overloading.	10	L3	CO1
OR					
Q.6	a.	Discuss the following terms in detail with example :	10	L2	CO3
		(i) static			
		(ii) final			
		(iii) finalize ()			

	b.	Define function overloading. Develop a program to define three overloaded function to find sum of two integer and sum of two floating point number.	10	L3	CO3
Module – 4					
Q.7	a.	Define inheritance. How super key is used in inheritance? Explain two general form of super key.	10	L1	CO3
	b.	Explain the concept of method overriding in JAVA programming with example.	10	L3	CO3
OR					
Q.8	a.	Illustrate with example a super class variable reference a subclass object.	10	L2	CO3
	b.	Explain multilevel hierarchy with an example program. Give the usage of final with inheritance.	10	L2	CO3
Module – 5					
Q.9	a.	Describe the various level of access protection available for packages and their implications.	10	L2	CO4
	b.	Give the basic form of an exception handling block.	5	L1	CO4
	c.	What is the importance of the clause finally?	5	L1	CO4
OR					
Q.10	a.	Define exception. Write a JAVA program which contain one method which will throw Illegal Access Exception and use proper exception handler so that exception should be printed.	8	L3	CO4
	b.	Define Package. What are the steps involved in creating user defined package with an example.	8	L3	CO4
	c.	Explain chained exception in JAVA.	4	L2	CO4

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BPOPS103/203

**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Principles of Programming Using C

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Discuss the structure of 'C' program with an example.		8	L2	CO1
	b.	What are Variables? Write the rules to declare a variable.		6	L2	CO2
	c.	What are escape sequences? Mention the escape sequences of 'C' language with their meaning.		6	L2	CO2
OR						
Q.2	a.	Explain any two output devices in detail.		6	L2	CO1
	b.	With an example, explain Input – Output statements in 'C'.		6	L2	CO2
	c.	Discuss the classification of Computers.		8	L2	CO1
Module – 2						
Q.3	a.	List all decision control statements in 'C'. Explain else – if ladder and nested if with its syntax and example.		8	L2	CO2
	b.	Write a 'C' program to simulate a calculator using switch statement.		6	L3	CO2
	c.	Explain break and continue statement with example.		6	L3	CO2
OR						
Q.4	a.	Explain the for Loop with its syntax. Write a 'C' program to find whether a given number is prime or not.		8	L3	CO2
	b.	Differentiate while and do – while loops with example.		6	L3	CO2
	c.	What are nested loops? Give example. Write a 'C' program to display the pattern shown below. 1 2 2 3 3 3 4 4 4 4 5 5 5 5 5		6	L3	CO2
Module – 3						
Q.5	a.	Define Function. Write the syntax of a function. Explain the categories of function with examples.		10	L2	CO2

	b.	Briefly explain the storage classes supported by 'C' language.	10	L2	CO2
OR					
Q.6	a.	Differentiate pass by value and pass by address parameter passing techniques.	5	L3	CO3
	b.	How 2 – dimensional arrays are declared and initialized? Write a 'C' program to find the transpose of a matrix.	7	L4	CO3
	c.	Define Recursion. Mention the properties of Recursion function. Write a 'C' program to find GCD of 2 numbers using recursive function.	8	L4	CO3
Module – 4					
Q.7	a.	What are Strings? Explain the 'C' function used to read and write characters.	6	L2	CO3
	b.	Write a program to find length of given string without using built in function.	6	L3	CO2
	c.	What is a Pointer? How pointer are declared and initialized? Mention the various operations that are carried out on pointers.	8	L4	CO4
OR					
Q.8	a.	Write a program to copy and concatenate from one string to another.	8	L3	CO5
	b.	Explain any 6 string manipulation functions.	6	L3	CO4
	c.	Write a 'C' program to find sum, mean, standard deviation of all elements in an array using pointers.	6	L5	CO5
Module – 5					
Q.9	a.	What is Union? Give its syntax. Differentiate unions and structures.	8	L2	CO4
	b.	Explain with an example array of structures and arrays within structure.	6	L2	CO4
	c.	Write a note on Structures and Functions.	6	L3	CO4
OR					
Q.10	a.	What is a File? Explain different modes of File with example.	8	L2	CO5
	b.	Write a note on Enumerated Data type.	6	L2	CO4
	c.	Write a 'C' program to copy the contents from one file to another.	6	L4	CO5

CBCS SCHEME

BPWSK106/206

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Question Paper Version : A

**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Professional Writing Skills in English

Time: 1 hrs.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners on the OMR sheets are strictly prohibited.**

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1. Identify the sentence with a parts of speech error:
a) She plays the piano very well b) They are listening to music
c) He are playing soccer d) We is going to the movies.
 2. Which sentence contains a parts of speech error?
a) He are a good student. b) She sings beautifully.
c) We will eat pizza for dinner. d) They are playing soccer outside.
 3. In which sentence is the error in parts of speech?
a) She is reading an interesting book. b) I can swimming very well.
c) They are watching a movie. d) He plays the guitar
 4. What is the grammatical term for a word that connects words, phrases, or clauses?
a) Conjunction b) Preposition c) Adverb d) Interjection
 5. In the sentence, "I saw him running down the street, "what is the function of the word "running"?
a) Adjective b) Adverb c) Gerund d) Infinitive
 6. What is the function of a participle in a sentence?
a) To modify a noun or pronoun b) To express an action
c) To function as the subject of a verb d) To function as the object of a verb
 7. In the sentence, "The book on the table is mine," what is the function of the phrase "on the table"?
a) Adjective b) Adverb
c) Prepositional phrase d) Gerund phrase

8. Which sentence has a grammatical error?
 - a) He enjoy playing basketball.
 - b) They is going to the beach.
 - c) She have finished her homework.
 - d) We am going to the park.
9. What is the grammatical mistake in the sentence: "The dog laying in the sun"?
 - a) Verb tense
 - b) Subject-verb agreement
 - c) Verb form
 - d) Word choice
10. What is the grammatical mistake in the sentence: "The car is blue, and the truck is red color"?
 - a) Run-on sentence
 - b) Comma splice
 - c) Redundancy
 - d) Fragment sentence
11. Which sentence contains a grammatical error?
 - a) The cat is sleeping on the couch.
 - b) She are a doctor.
 - c) They is going to the park.
 - d) I like to sing.
12. What is the grammatical error in the sentence: "He don't have any money"?
 - a) Subject-verb agreement
 - b) Pronoun usage
 - c) Tense consistency
 - d) Sentence structure
13. Which of the following sentences has a grammatical error?
 - a) She sings beautifully.
 - b) We am going to the park.
 - c) They are watching a movie.
 - d) He plays basketball every weekend.
14. Which sentence demonstrates correct subject-verb agreement?
 - a) The dogs barks loudly in the park.
 - b) The dog bark loudly in the park.
 - c) The dogs bark loudly in the park.
 - d) The dog barks loudly in the park.
15. Which option demonstrates correct subject-verb agreement?
 - a) The team are playing well this season.
 - b) The team is playing well this season.
 - c) The team were playing well this season.
 - d) The team am playing well this season.
16. Which of the following sentences is in the present perfect tense?
 - a) She will go to the store.
 - b) She is going to the store.
 - c) She went to the store.
 - d) She has gone to the store.
17. What is the future tense of the verb "to swim"?
 - a) Swim
 - b) Swam
 - c) Swimming
 - d) Will swim
18. Which tense is used to describe actions that are ongoing or happening at the moment?
 - a) Present simple
 - b) Past simple
 - c) Present continuous
 - d) Past continuous
19. What is the past tense of the verb "to eat"?
 - a) Eating
 - b) Eaten
 - c) Ate
 - d) Eat
20. In the sentence, "I have lived here for ten years," which tense is used for the verb "have lived"?
 - a) Present simple
 - b) Present perfect
 - c) Past simple
 - d) Past perfect

21. What is the future tense of the verb "to run"?
a) Runs b) Running c) Ran d) Will run
22. Which of the following sentences is in the past continuous tense?
a) She has been studying all day. b) She studied for the exam last night.
c) She was studying when the phone rang. d) She will study tomorrow.
23. Which of the following refers to words that have similar meanings?
a) Synonyms b) Antonyms c) Homonyms d) Homophones
24. What term describes words that are spelled the same but have different meanings and sometimes different pronunciations?
a) Synonyms b) Antonyms c) Homonyms d) Homophones
25. Complete the analogy: Cat is to kitten as dog is to _____.
a) Cub b) Puppy c) Foal d) Chick
26. Choose the correct usage: "Yours" versus "You're".
a) Your going to love the party. b) You're going to love the party.
c) Youre going to love the party. d) You are going to love the party.
27. Choose the correct usage: "Its" versus "It's".
a) Its a beautiful day outside. b) It's a beautiful day outside.
c) Its raining heavily d) It is raining heavily.
28. Choose the correct collocation:
a) Make an appointment. b) Do an appointment.
c) Take an appointment. d) Give an appointment.
29. Identify the correct collocation:
a) Take a decision b) Make a decision
c) Do a decision d) Give a decision
30. Which contraction is correct for "they have"?
a) they've b) they're c) they'd d) they'll
31. Choose the correct contraction for "we will":
a) we're b) we've c) we'd d) we'll
32. What is the contraction of "cannot"?
a) can't b) couldn't c) can've d) can't've
33. Rearrange the words to form a grammatically correct sentence "Yesterday, went, to, I, store, the."
a) Yesterday I went to store the. b) Went I to the store yesterday.
c) Yesterday I went to the store. d) I went to the store yesterday.
34. Rearrange the words to form a coherent sentence: "She books, enjoys, reading, many, by, written, different, authors."
a) She enjoys reading many books by different authors written.
b) She enjoys readings by many books written different authors.
c) She enjoys reading many books written by different authors.
d) She by different enjoys readings many books written authors.

35. Rearrange the words to form a grammatically correct sentence: "In, I, morning, the go, always, for, a, run."
a) Always in the morning go I for a run.
b) In the morning always I go for a run.
c) I always go for a run in the morning.
d) In the always morning I go for a run.
36. Rearrange the words to form a coherent sentence: "Hiking, in, we, enjoyed, the, mountains, beautiful, the."
a) We enjoyed hiking in the mountains the beautiful.
b) In the mountains we enjoyed hiking the beautiful.
c) We enjoyed the beautiful mountains in the hiking.
d) The beautiful mountains in the hiking we enjoyed.
37. Identify the error in the sentence: "The dog wagged it's tail excitedly."
a) The b) dog c) wagged d) it's
38. What is the active voice form of the sentence: "The cake was baked by Sarah"?
a) Sarah has baked the cake. b) Sarah had baked the cake.
c) Sarah bakes the cake. d) Sarah will bake the cake.
39. Which option correctly transforms the sentence into passive voice: "The teacher teaches the students"?
a) The teacher is teaching by the students.
b) The students are being taught by the teacher.
c) The teacher was teaching the students.
d) The students were taught by the teacher.
40. What is the reported speech form of the direct speech: "He said, 'I am leaving for London tonight'?"
a) He said that he leaves for London tonight.
b) He said that he left for London tonight.
c) He says that he leaves for London tonight.
d) He says that he left for London tonight.
41. Choose the correct transformation of the sentence into reported speech: "She said, 'I have already finished my homework.'"
a) She said that she already finished her homework.
b) She says that she has already finished her homework.
c) She said that she has already finished her homework.
d) She says that she had already finished her homework.
42. Identify the error in the sentence: "He don't have enough money to buy a car."
a) He b) don't c) have d) enough
43. Which part of the sentence contains the error: "The student's scored high marks in the exam."
a) The b) student's c) scored high marks d) in the exam
44. Find the error in the sentence: "She sings beautiful."
a) She b) sings c) beautiful d) no error

45. Where is the error in the sentence: "Their going to the beach tomorrow."
a) Their b) going c) to the beach d) tomorrow
46. Identify the error in the sentence: "The book is laying on the table."
a) The b) book c) is laying d) on the table
47. Which option improves the sentence: "He plays good football."
a) He plays well football. b) He plays football good.
c) He plays football well. d) He plays football goodly.
48. Choose the improved version of the sentence "The weather is too much hot today."
a) The weather is very hot today. b) The weather is too hot today.
c) The weather is much hot today. d) The weather is hot too much today.
49. Which option corrects the sentence: "I am not feeling good today."
a) I am not feeling well today. b) I am not feeling goodly today.
c) I am not feeling better today. d) I am not feeling good today.
50. Choose the corrected version of the sentence: "He goes to gym regular."
a) He goes to gym regularly. b) He goes to the gym regular.
c) He goes to the gym regularly. d) He goes to gym regulars.

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